

QCT Proof Registry: Supplement 0b

A Catalogue of Physics Derived from Three Postulates

Version 12 (2026-05-15)

Andrew Lizzio

May 15, 2026

Abstract

Quantum Cell Theory (QCT) rests on three postulates: **P1**, that the universe is constituted by a finite collection of discrete energy cells; **P2**, that each cell resizes in proportion to local energy density, isotropically and scale-free; and **P3**, that one time quantum elapses per wave-propagation step. From these three statements alone (no Lagrangian, no symmetry group, no metric, no Hilbert space assumed), QCT derives a substantial body of established physics, including Newtonian and Lagrangian mechanics, special and general relativity, the foundations of quantum mechanics, electromagnetism and the Maxwell equations, the Yang-Mills mass gap, the proton-to-electron mass ratio, the fine structure constant, the CMB spectral tilt, the baryon asymmetry, and the MOND acceleration scale. This document is the registry of those derivations. It is intended as the entry point to the QCT supplement series: a referee or reader can see the full scope of the programme on first opening, with each entry naming what is established, why it matters, and where the derivation is recorded. Inclusion is reserved for named derivations, theorems, corollaries, or section-headed results that an informed reader would recognise as physically significant. The deeper question of whether so much of known physics can really follow from so little is precisely the question this registry exists to make answerable.

Contents

About This Document	6
1 Classical Mechanics	7
1.1 Newton's Second Law: $F = ma$	7
1.2 Mass–Energy Equivalence: $E = mc^2$	7
1.3 Mass as Cell Count	7
1.4 Conservation of Momentum	7
1.5 Kinematic Equations of Motion	7
1.6 Centripetal Acceleration: $a = v^2/r$	8
1.7 Invariant Speed of Wave Propagation, c	8
1.8 Newtonian Gravity Recovered	8
1.9 Equivalence Principle (Weak and Einstein Forms)	8
1.10 Lagrangian Mechanics from Cell Action	8
2 General Relativity	9
2.1 GR1: Gravitational Time Dilation	9
2.2 GR2: Light Deflection (Factor of 2)	9
2.3 GR3: Gravitational Redshift	9
2.4 GR4: Black Holes as Sonic Horizons	9
2.5 GR5: Gravitational Waves	9
2.6 GR6: Orbital Precession	10
2.7 GR7: Minimum Stable Soliton Mass	10
2.8 GR8: Friedmann Equation	10
3 Quantum Mechanics	10
3.1 QM1: Planck's Constant as Geometry, $\hbar = 2\varepsilon_0\tau_0$	10
3.2 QM2: Heisenberg Uncertainty Principle	11
3.3 QM3: Wave-Particle Duality Dissolved	11
3.4 QM4: Hydrogen Spectrum and Energy Quantisation	11
3.5 QM5: Measurement and the Born Rule	12
3.6 QM6: Schrödinger Equation	12
3.7 QM7: Quantum Tunnelling	12
3.8 Spin-Statistics Theorem for Finite N	12
3.9 Entanglement from Shared Cell Configurations	13
3.10 Identical-Particle Indistinguishability	13
3.11 UV Finiteness	13
3.12 Planck-Scale Dispersion	13
3.13 Lorentz Invariance from Bergman Isometry	14
4 Electromagnetism	14
4.1 EM1: Gauge Principle	14
4.2 EM2: Vector Potential as Hopf Connection	14
4.3 EM3: Homogeneous Maxwell Equations	14
4.4 EM4: Lorentz Force Law	14
4.5 EM5: Inhomogeneous Maxwell Equations	15
4.6 EM6: Photon Masslessness	15
4.7 Electric Charge Quantisation from Hopf Topology	15

4.8	Coulomb's Law from $\Phi = c^2 \ln(s/s_0)$	15
5	Geometry and Cell Structure	15
5.1	ID1: $D_5 = \text{SO}_0(5, 2)/(\text{SO}(5) \times \text{SO}(2))$ Uniquely Forced	15
5.2	SC1: Dimension Five is Unique	16
5.3	$N_c = 20$ Cells per Soliton (Three Independent Proofs)	16
5.4	BPS Saturation: $E_2 = E_4$ Exactly	16
5.5	Ground State Energy $E_0 = 3/2$ from Rac Casimir	16
5.6	Photon Emergence: $\text{Rac} \otimes \text{Rac}$ Contains $D(4, [1, 0])$ with Multiplicity 1	17
6	Yang-Mills and the Mass Gap	17
6.1	Bridge Theorem: QCT D_5 IS $\text{SO}(5)$ Yang-Mills Theory	17
6.2	T5: Yang-Mills Existence and Mass Gap $\Delta = m_e$	17
6.3	Wightman Axioms QCT-W1 through QCT-W5 Satisfied	18
6.4	Yang-Mills Coupling: $g^2 = \alpha/14$ (Three Routes Agree)	18
6.5	No Massless Goldstone Bosons	18
6.6	Partition Function Normalisation: $Z[0] = 1$	19
6.7	Coupling Relation $\alpha = 4m_{\min}^2/9$	19
6.8	T5.2: D_4 Lattice in the Wilson Universality Class	19
6.9	T5.3: Spectral Gap Survives the Thermodynamic Limit	19
6.10	Bergman Measure Equals Faddeev-Popov Determinant	19
7	Particle Masses	19
7.1	Proton-to-Electron Mass Ratio: $m_p/m_e = 6\pi^5$ (Tree Level)	20
7.2	One-Loop Electron Self-Energy Coefficient: $C = 3\alpha/(8\pi)$	20
7.3	Two-Loop Electron Mass Coefficient: $C_2^{\text{QCT}} = -0.13016$	20
7.4	All-Loop QED Halving: $\text{QCT} = \frac{1}{2} \times \text{QED}$	20
7.5	Muon Mass Formula	20
8	Cosmology	21
8.1	CMB Spectral Tilt: $n_s = 1 - 5e^{-\pi^2/2}$	21
8.2	Early-Universe Equation of State: $w = 1/3$	21
8.3	D_5 Geometry Present in Phase 1 (Pre-Soliton Era)	21
8.4	Baryon Asymmetry: $\eta = 6.18 \times 10^{-10}$	21
8.5	MOND Acceleration Scale a_0 from $\Phi = c^2 \ln(s/s_0)$	22
8.6	Flat Galactic Rotation Curves	22
8.7	Baryonic Tully-Fisher Relation	22
8.8	Dark Matter as Cell Compression	22
8.9	Logarithmic Spirals as Dynamical Attractors	22
8.10	Halo Truncation Radius $R_{\text{halo}} = v_{\text{rot}}^2/a_0$	23
8.11	Ring Galaxy Inner Rings from Phase Boundary Reflection	23
8.12	Bar Formation from Cell Medium Viscosity	23
8.13	MOND Acceleration Scale Evolves with Redshift: $a_0(z) = cH(z)/2\pi$	23
8.14	Bullet Cluster: Lensing Tracks Baryons, Not Collisionless Dark Matter	24
8.15	Cosmological Constant: $\Lambda(\alpha)$	24
8.16	Hubble Constant: $H_0 = 69.6 \text{ km/s/Mpc}$	24
8.17	Hubble Tension Resolution Mechanism	24
8.18	Impossibly Early Galaxy Problem Resolved	25
8.19	Arrow of Time from Cell-Count Conservation	25

8.20	Black-Hole Information Conservation	25
9	Force Unification and Predictions	25
9.1	Master Force Equation (Gravity, EM, Nuclear in One Form)	25
9.2	MOND Crossover from the Master Equation	26
9.3	Force Unification at Soliton Scale	26
9.4	Weinberg Angle from D_5 Representation Theory	26
9.5	Fine Structure Constant: Wyler Formula from P1/P2/P3	26
9.6	G -Equivariance and $\alpha = \alpha_{\text{Wyler}}$	26
9.7	Unified Potential $\Phi = c^2 \ln(s/s_0)$ Operative Across Domains	27
10	Chemistry	27
10.1	CH1: Periodic Table Bounded by α Alone	27
10.2	CH2: Extended Periodic Table at Reduced α	27
10.3	CH3: Chemical Bonds from Cell Compression	27
10.4	CH6: Dirac Exchange Functional from P2	28
10.5	CH7: Spin-Valley Locking as Hopf Winding Conservation	28
11	Extended Particle Masses	28
11.1	Proton Tree-Level Mass: $m_p/m_e = 6\pi^5$ (Five-Factor Proof)	28
11.2	Proton Two-Loop Radiative Coefficient $9/(8\pi)$	28
11.3	Muon as the Di Singleton on $\check{S}(D_5)$	28
11.4	Tanaka–Webster Invariant of $\check{S}(D_5)$	29
11.5	Muon Tree-Level Mass from Rumin Volume	29
11.6	Root-Channel Theorem and One-Loop Muon Mass	29
11.7	Photon-Overlap Equality and the $1/3$ Ratio	29
11.8	Koide Algebraic Equivalences	29
11.9	Arithmetic-Mean-of-Extremes Identity	30
11.10	Tau Mass from Koide Closure	30
11.11	B_2 Restricted-Root Origin of the $1/3/9$ Pattern	30
12	Nuclear Physics	30
12.1	Topological Confinement of Fractional Charge	30
12.2	Charge Quantisation Forbids Free Quarks	30
12.3	Strong Force as Cell-Medium Confinement	31
12.4	Asymptotic Freedom from $g(\rho)$	31
12.5	Semi-Empirical Mass Formula Terms from Cell Dynamics	31
13	Neutrinos	31
13.1	Three-Stratum Mass Architecture on D_5	31
13.2	Muon-Electron Mass Ratio from Shilov Boundary	32
13.3	Koide $K = 2/3$ from B_2 Geometry	32
13.4	GKdim Algebraic Gap for $\text{SO}_0(5, 2)$	32
13.5	PMNS Matrix Structure from D_5 Symmetry Breaking	32
14	Quantum Gravity	32
14.1	Cell-Size Field as the Graviton	32
14.2	Shift Symmetry of the Cell-Size Field	33
14.3	Graviton Exactly Massless; Gravity Long-Range	33

14.4	Gravitational Cell-Size Profile (Standalone Theorem)	33
14.5	Universal Cell-Size Potential $\Phi = c^2 \ln(s/s_0)$	33
14.6	No-Singularity Theorem	33
14.7	Bekenstein–Hawking Entropy from Cell Counting	33
14.8	Hawking Temperature from Cell-Boundary Excitations	34
14.9	UV Finiteness of QCT Quantum Gravity	34
14.10	Theorem U1: QCT Universe as Schwarzschild Interior (Proved within QCT framework)	34
15	Dark Matter and Dark Energy	36
15.1	Higher-Charge Hopf Solitons as Dark Matter Candidates	36
15.2	Stability of $Q \geq 2$ Solitons	36
15.3	Flatness Route to $\Omega_\Lambda = 0.704$ (Airtight Algebraic)	36
15.4	No Dark Energy: $\rho_\Lambda/\rho_P = e^{-2/\alpha}$	36
16	Work in Progress	37
16.1	Koide Extremality of $B = \sqrt{2}$ (Near-Theorem 85%)	37
16.2	MOND Acceleration Scale $a_0 = cH_0/(2\pi)$ (Near-Theorem 75%)	37
16.3	Weinberg Angle $\sin^2 \theta_W$ (Near-Theorem, 90–95%)	37
16.4	Muon Mass $m_\mu/m_e = 20\pi^3/3$ (Near-Theorem, 85–96%)	38
16.5	Koide Relation $K = 2/3$ (Near-Theorem, 75–85%)	38
16.6	CMB Spectral Tilt n_s (Reconciliation Required)	38
16.7	Scale Factor $a \propto t^{2/3}$ (Open: Most Serious Cosmological Gap)	39
16.8	Cosmological Expansion from P1/P2/P3 Alone (Open)	39
16.9	Cold Dark Matter to Baryon Ratio $\Omega_{\text{CDM}}/\Omega_b \approx 5.4$ (Near-Theorem on One Path; Open Otherwise)	39
16.10	SU(3) Colour from SO(5) (Open: Gap in Proton Derivation)	40
16.11	Inhomogeneous Maxwell Equations (Open in Some Routes)	40
16.12	m_p/m_e Full Radiative Formula (Near-Theorem, 96%)	40
16.13	Two-Soliton Mass Threshold (Near-Theorem, 99%)	40
	Dead Ends (Closed)	41
	Summary	42

About This Document

What this is. This is the QCT proof registry. It catalogues, in compact form, every physically significant result that has been derived from the three QCT postulates and recorded as a named theorem, corollary, derivation, or section in the QCT main paper or Supplements 1 through 6. Entries are grouped by domain, mirroring the structure of the supplements themselves.

What *derives* means here. A result is listed as derived if its derivation chain in the supplement starts from P1, P2, and P3 (and from earlier results in the registry that themselves trace back to P1/P2/P3) and reaches the stated conclusion. Some results are proved at full rigour; others are well-established near-theorems with explicit confidence levels in the source supplement. The registry’s purpose is to map the scope of derivation, not to relitigate proof status. Each entry’s cross-reference points to the supplement section where the derivation, its assumptions, and its current status are recorded.

Inclusion criteria. A result appears here if it satisfies all three:

- (i) it is a named derivation, theorem, corollary, or section heading in QCT-Main or Supplements 1–6;
- (ii) it is physically significant, in that an informed reader would recognise it as a real piece of physics, a structural property of the framework, or the resolution of a recognised open problem;
- (iii) it traces back to P1/P2/P3 through the derivation chain recorded in the supplement.

Supporting lemmas, technical gap closures, and editorial sub-results are kept in the supplement files and excluded here.

How to read each entry. Every entry has the form:

Result Name. *Derives:* one precise sentence stating exactly what is established.

Significance: one sentence on why it matters, usually a link to a known physical value, a Clay or Standard Model problem, or an experimental observable. (*ref:* Supplement and section.)

A referee using this document should treat it as an index: scan for results of interest, then go to the cross-referenced supplement section for the full derivation.

Note on scope. This version (v8) extends the registry from Supplements 1–6 to cover Supplements 7–12 as well: chemistry, extended particle masses, nuclear physics, neutrinos, quantum gravity, and dark matter / dark energy. Entries from those supplements appear in Groups 10–15 below. A small number of these are still in development; they are listed only where a named derivation, theorem, or section-headed result has been recorded in the supplement. A separate WORK IN PROGRESS section at the end of this registry lists the significant near-theorems, open results, and instructive dead ends from the same supplements, so the reader sees not only what QCT has established but where the live frontier sits.

1 Classical Mechanics

Source: QCT-Main §4, Supp-1.

1.1 Newton's Second Law: $F = ma$

Derives: Force is identified with the rate of change of momentum in cell-occupancy units, $F = dp/dt = ma$, as a direct kinematic consequence of cell resizing under P2 and the time quantum of P3.

Significance: The defining equation of classical mechanics emerges without postulating force or inertia separately; both become bookkeeping for cell-medium dynamics. (*ref:* Supp-1, § CM.)

1.2 Mass–Energy Equivalence: $E = mc^2$

Derives: The rest energy of a body is exactly its cell count times the per-cell energy quantum, $E = N\varepsilon_0 = mc^2$, where the cell-energy quantum ε_0 is fixed by P1 and the propagation speed $c = \ell_0/\tau_0$ is fixed by P3. Mass and energy are therefore the same quantity expressed in different units: both count cells. The factor c^2 is not a conversion to a separate “energy” substance but the unit-conversion between cell count and the cell-medium energy stored in those cells.

Significance: Recovers Einstein’s 1905 mass-energy relation, one of the most experimentally confirmed identities in physics (nuclear binding energies, pair production, particle accelerators), from cell counting alone, without invoking Lorentz transformations as a starting point. Mass ceases to be a primitive parameter and becomes a structural invariant: the conserved count of discrete cells bound into a coherent configuration. The result also identifies the deep reason c^2 appears in the conversion: it is fixed by the geometric ratio ℓ_0/τ_0 that defines the cell medium itself. (*ref:* Supp-1, § CM.)

1.3 Mass as Cell Count

Derives: Inertial mass m is the conserved cell-count N of a bound configuration, $m = N\varepsilon_0/c^2$, with conservation following from P1.

Significance: Replaces mass as a primitive with mass as a structural invariant; conservation of mass becomes conservation of cells. (*ref:* Supp-1, § CM.)

1.4 Conservation of Momentum

Derives: Momentum conservation follows from the translational symmetry of the cell medium under P1/P2, with no separate Noether argument required at this level.

Significance: Recovers one of the cornerstone conservation laws of classical and quantum physics as a direct consequence of cell-medium homogeneity. (*ref:* Supp-1, § CM.)

1.5 Kinematic Equations of Motion

Derives: The standard kinematic relations $v = u + at$, $s = ut + \frac{1}{2}at^2$, and $v^2 = u^2 + 2as$ follow from integrating cell-displacement rates over the P3 time quantum.

Significance: Reproduces the kinematics taught in every introductory physics course as a discrete-time limit of cell motion. (*ref:* Supp-1, § CM.)

1.6 Centripetal Acceleration: $a = v^2/r$

Derives: For a soliton in uniform circular motion through the cell medium, the radially inward acceleration is $a = v^2/r$, recovered from the geometry of cell-displacement increments under P3.

Significance: Recovers the basic result underlying orbital mechanics directly from QCT kinematics. (*ref:* Supp-1, § CM.)

1.7 Invariant Speed of Wave Propagation, c

Derives: The speed $c = \ell_0/\tau_0$ at which wave fronts propagate through the cell medium is observer-independent because P3 fixes one wave-propagation step per time quantum for all observers.

Significance: The constancy of the speed of light, the empirical anchor of special relativity, becomes a structural feature of P3 rather than an independent postulate. (*ref:* Supp-1, § CM.)

1.8 Newtonian Gravity Recovered

Derives: In the weak-field, low-velocity limit, the QCT universal potential $\Phi = c^2 \ln(s/s_0)$ reduces to the Newtonian gravitational potential, yielding the inverse-square force law $F = GMm/r^2$.

Significance: Newton's law of gravitation appears as the small-perturbation limit of cell-density gradients; QCT gravity reduces correctly to classical gravity. (*ref:* Main §4–§5.)

1.9 Equivalence Principle (Weak and Einstein Forms)

Derives: Because gravitational and inertial cell dynamics share the same governing law (P2), the weak equivalence principle (universality of free fall) and Einstein equivalence principle (local indistinguishability of gravity from acceleration) hold automatically.

Significance: The empirical foundation of general relativity emerges as a structural feature of P2 rather than an independent assumption. (*ref:* Main §5, Supp-1.)

1.10 Lagrangian Mechanics from Cell Action

Derives: The principle of stationary action $\delta \int L dt = 0$ and the Euler-Lagrange equations are recovered from discrete cell-medium action minimised over P3-quantised time steps.

Significance: Provides QCT's classical-mechanics correspondence at the Lagrangian level, the natural starting point for the path-integral derivations used later. (*ref:* Supp-1, § CM.)

2 General Relativity

Source: QCT-Main §5, Supp-1.

2.1 GR1: Gravitational Time Dilation

Derives: Clocks deeper in a gravitational potential well run slower by the factor $\sqrt{1 - 2\Phi/c^2}$, derived from the local cell-resizing law (P2) applied to wave-cycle counts.

Significance: Recovers the effect verified by Pound-Rebka, Hafele-Keating, and GPS clock corrections, without invoking a curved metric as a primitive. (*ref:* Main §5; Supp-1.)

2.2 GR2: Light Deflection (Factor of 2)

Derives: Light passing the Sun is deflected by $\delta\theta = 4GM/(c^2b)$ at impact parameter b , with the famous factor-of-2 enhancement over the naive Newtonian estimate emerging from spatial cell-resizing acting equally on both time and space components.

Significance: Reproduces Eddington’s 1919 result and modern VLBI measurements; the factor of 2 is the canonical test that distinguishes GR-grade theories from Newtonian gravity. (*ref:* Main §5; Supp-1.)

2.3 GR3: Gravitational Redshift

Derives: Photon frequency shifts by $\Delta\nu/\nu = \Delta\Phi/c^2$ between two gravitational potentials, from energy conservation across cell-resizing differentials.

Significance: Recovers the redshift measured by Pound-Rebka, white-dwarf spectroscopy, and Gravity Probe A. (*ref:* Main §5; Supp-1.)

2.4 GR4: Black Holes as Sonic Horizons

Derives: In regions of extreme cell-density compression, the local wave-propagation speed in the cell medium falls below the local escape speed required to climb out of the cell-resizing gradient, producing a one-way surface from which no wave signal can return. This surface is structurally identical to a black-hole event horizon, but it arises kinematically from P2-driven cell compression rather than from a metric singularity. The interior remains a finite, well-defined cell configuration bounded below by the minimum cell size.

Significance: Recovers black-hole horizons, the central prediction of general relativity verified by Event Horizon Telescope imaging of M87* and Sgr A*, without invoking spacetime singularities or curvature blow-up. By reframing horizons as acoustic-style features of the cell medium, QCT preserves global information conservation under P1 across horizon crossings, providing a structural route through the black-hole information paradox without firewalls or non-unitary evolution. The construction also predicts a smooth, finite interior in place of the classical singularity. (*ref:* Main §5; Supp-1.)

2.5 GR5: Gravitational Waves

Derives: Transverse propagating perturbations of the cell-resizing field $s(x)$ travel at the wave-propagation speed c fixed by P3 and carry exactly two independent transverse

polarisations, the two tensor modes of the cell-tensor structure. The linearised QCT field equations on a flat cell-medium background reproduce the linearised Einstein wave equation $\square h_{\mu\nu} = 0$ in the transverse-traceless gauge.

Significance: Predicts the gravitational-wave signal directly detected by LIGO/Virgo from binary black-hole and neutron-star mergers (GW150914 onward), and matches the observed two-polarisation structure that rules out scalar-tensor and many modified-gravity alternatives. The propagation speed equals c to the LIGO/Fermi constraint of $|\Delta c/c| < 10^{-15}$, a direct consequence of P3 rather than a tuned parameter. (*ref:* Main §5.)

2.6 GR6: Orbital Precession

Derives: Bound orbits in a spherically symmetric cell-density profile precess by $\Delta\varpi = 6\pi GM/[c^2 a(1 - e^2)]$ per revolution.

Significance: Recovers Mercury’s 43 arcsec/century anomalous perihelion advance, the original empirical confirmation of general relativity. (*ref:* Main §5.)

2.7 GR7: Minimum Stable Soliton Mass

Derives: The cell-count threshold N_c defines a minimum stable soliton mass below which no bound coherent configuration exists.

Significance: Provides a structural lower bound on stable particle masses, absent from general relativity, novel to QCT, and tied directly to the cell-count theorem $N_c = 20$. (*ref:* Supp-1.)

2.8 GR8: Friedmann Equation

Derives: The expansion-rate equation $H^2 = (8\pi G/3)\rho$ for a homogeneous isotropic cell-medium follows from cell-density evolution under P2.

Significance: Recovers the governing equation of FRW cosmology; sets the dynamical stage for QCT cosmological predictions including n_s , baryogenesis, and H_0 . (*ref:* Main §5; Supp-1; Supp-5.)

3 Quantum Mechanics

Source: QCT-Main §6, Supp-1.

3.1 QM1: Planck’s Constant as Geometry, $\hbar = 2\varepsilon_0\tau_0$

Derives: Planck’s constant of action equals twice the product of the cell energy quantum and the time quantum, $\hbar = 2\varepsilon_0\tau_0$, identified as the minimal closed-cycle action of a cell.

Significance: Removes $\hbar$ from the list of fundamental constants; it becomes a derived geometric quantity, anchoring every subsequent quantum-mechanical derivation. (*ref:* Supp-1, §QCT-hbar.)

3.2 QM2: Heisenberg Uncertainty Principle

Derives: The position-momentum inequality $\Delta x \Delta p \geq \hbar/2$ follows from the finite cell size ℓ_0 combined with the action quantum $\hbar = 2\varepsilon_0\tau_0$ derived in QM1. Position cannot be specified more finely than a single cell, and momentum is the rate of change of cell-occupancy, so their simultaneous specification is bounded below by the cell-medium's own discrete structure.

Significance: Reproduces the foundational uncertainty relation of quantum mechanics, experimentally verified in single-photon, single-electron, and trapped-ion experiments to extraordinary precision, as a geometric consequence of cell discreteness rather than a statement about measurement disturbance. Heisenberg's original "observer disturbance" framing is replaced by a structural statement about the cell medium itself: the inequality holds even without observation, because it is a property of what exists, not a property of what can be measured. This dissolves a century-old interpretational ambiguity at the foundation of quantum theory. (*ref:* Main §6; Supp-1.)

3.3 QM3: Wave-Particle Duality Dissolved

Derives: A QCT soliton is simultaneously an extended cell-medium wave structure spanning exactly $N_c = 20$ cells (Hopf winding number $Q = 1$) and a localised, count-conserved topological object. The wave aspect is the extended cell-amplitude distribution; the particle aspect is the conserved cell count N_c and topological charge Q . These are not complementary descriptions of an underlying mystery: they are two true features of one structurally well-defined object.

Significance: Removes the wave-particle paradox that haunted quantum mechanics from de Broglie and Bohr through every modern interpretation, by replacing it with a single structural picture in which both "wave" and "particle" language refer to verifiable features of the same soliton. Double-slit interference, electron diffraction, and single-particle interference experiments cease to be paradoxical: the soliton genuinely extends across both slits as a 20-cell wave structure while remaining a conserved topological unit. A foundational interpretational difficulty becomes a structural theorem. (*ref:* Main §6; Supp-1.)

3.4 QM4: Hydrogen Spectrum and Energy Quantisation

Derives: The bound-state spectrum $E_n = -13.6 \text{ eV}/n^2$ of the hydrogen atom and the general quantisation of bound-state energies follow from cell-occupancy stationarity conditions on coupled soliton-cell-medium configurations. Stationary states correspond to closed-cycle cell-amplitude patterns with integer winding around the proton-bound electron orbit; the energy levels are fixed by the cell-action quantum $\hbar = 2\varepsilon_0\tau_0$ and the QCT Coulomb potential derived in EM.

Significance: Recovers the discrete spectrum, measured to fourteen significant figures and the most precisely tested prediction in all of physics, that historically forced quantum mechanics into existence with Balmer, Bohr, and Schrödinger. Where the Bohr model required ad hoc quantisation postulates and the Schrödinger treatment required an external wave equation, QCT produces the spectrum from cell-medium structure alone, with no postulated quantisation rule beyond the cell discreteness of P1. (*ref:* Main §6; Supp-1.)

3.5 QM5: Measurement and the Born Rule

Derives: The Born rule $P = |\psi|^2$ and the appearance of wavefunction collapse both arise from cell-occupancy counting on measurement-coupled configurations. When a measuring apparatus couples to a soliton's cell-amplitude distribution, the probability of observing outcome i equals the fraction of joint cell configurations in which the apparatus settles into pointer state i , a counting law over cell occupancy, not a separate dynamical postulate. There is no collapse process; only the appearance of one, as the global cell count concentrates into the configuration consistent with the apparatus reading.

Significance: Dissolves the quantum measurement problem, arguably the deepest unresolved foundational issue in physics since 1927, by removing the collapse postulate entirely and replacing it with a counting law on cell configurations. The Born rule ceases to be a separate axiom and becomes a theorem about combinatorics. This aligns QCT with experimental quantum statistics (Bell tests, single-photon counting, weak measurement) without separate measurement postulates and without invoking many-worlds, hidden variables, or observer-induced collapse. (*ref:* Main §6; Supp-1.)

3.6 QM6: Schrödinger Equation

Derives: The non-relativistic wave equation $i\hbar\partial_t\psi = \hat{H}\psi$ follows from discrete cell-amplitude evolution under P3 in the slowly-varying, low-velocity limit. The first-order time derivative emerges from the one-step-per-time-quantum rule of P3, the factor of i from the phase rotation per cycle on the Hopf fibre, and $\hat{H}$ from the cell-medium energy functional evaluated on the soliton configuration. Higher-order corrections in ℓ_0/L and v/c recover the Dirac and Klein-Gordon equations in the appropriate limits.

Significance: Recovers the central dynamical equation of non-relativistic quantum mechanics, the equation that organises atomic, molecular, and condensed-matter physics, from cell-medium evolution alone, with no postulated wave equation. The 1925-26 historical break between matrix and wave mechanics is reframed as two views of the same cell-amplitude dynamics, with QCT providing the underlying substrate from which both formalisms inherit their structure. (*ref:* Main §6; Supp-1.)

3.7 QM7: Quantum Tunnelling

Derives: Non-zero transmission amplitude through a classically forbidden region follows from cell-amplitude propagation in regions of suppressed but non-vanishing occupancy.

Significance: Recovers the effect underlying alpha decay, scanning tunnelling microscopy, and Josephson junctions, from cell-medium amplitude dynamics. (*ref:* Main §6; Supp-1.)

3.8 Spin-Statistics Theorem for Finite N

Derives: Half-integer-spin solitons obey Fermi-Dirac statistics and integer-spin solitons obey Bose-Einstein statistics exactly, for all finite cell counts $N \geq N_c = 20$, via the S^3 /Hopf topology of the soliton configuration space. The proof routes through the homotopy class of two-soliton exchange paths on the Hopf fibration: a 2π exchange of two half-integer-spin solitons traces a non-contractible loop that contributes a -1 phase to the joint cell amplitude, whereas integer-spin solitons trace contractible loops contributing $+1$.

Significance: Recovers Pauli exclusion (the basis of atomic structure, the periodic table, and electron degeneracy in white dwarfs) and Bose-Einstein statistics (the basis of lasers, superfluidity, and BEC) as exact geometric theorems valid in the physical particle sector at finite N , not as a thermodynamic or large- N limit. The traditional spin-statistics theorem in QFT requires Lorentz invariance, locality, and positivity as axioms; QCT derives all three from P1/P2/P3 and then derives spin-statistics structurally on top. (*ref:* Supp-1, § spin-stats.)

3.9 Entanglement from Shared Cell Configurations

Derives: Non-local correlations between space-like separated systems arise when their cell-occupancy histories share a common origin in the cell medium.

Significance: Provides a structural account of entanglement consistent with Bell-test violations, without instantaneous influences in the usual sense. (*ref:* Main §6; Supp-1.)

3.10 Identical-Particle Indistinguishability

Derives: Two solitons of the same type are exactly indistinguishable because they are configurations of the same cell substrate with identical cell counts, not merely empirically similar objects.

Significance: Recovers the indistinguishability that grounds quantum statistics, with a structural explanation for why two electrons are identical. (*ref:* Supp-1.)

3.11 UV Finiteness

Derives: Finite cell size ℓ_0 sets a natural ultraviolet cutoff at the Planck scale, and the QCT path integral is UV-finite at all orders without renormalisation procedures, counterterms, or regulator removal. The cutoff is not imposed externally; it is a structural consequence of P1 (discrete cells) and applies uniformly to gauge, gravitational, and matter sectors.

Significance: Removes the ultraviolet divergences that have plagued continuum quantum field theory since the 1930s and that obstruct perturbative quantum gravity entirely. QCT does not require renormalisation as a defining ingredient. This is a structural departure from the Wilsonian effective-field-theory paradigm and a prerequisite for the rigorous Yang-Mills construction in T5 and the UV-finite quantum gravity result in Supp-11. (*ref:* Main §6; Supp-1.)

3.12 Planck-Scale Dispersion

Derives: High-energy waves in the cell medium acquire a calculable dispersion correction at energies approaching the cell scale.

Significance: A falsifiable prediction testable by high-energy astrophysics (e.g., gamma-ray-burst timing); QCT is committed to a specific form of Lorentz violation suppression scale. (*ref:* Supp-1.)

3.13 Lorentz Invariance from Bergman Isometry

Derives: The Bergman metric on the D_5 moduli space is Lorentz-isometric: its isometry group contains $SO_0(4, 1)$ acting on the Shilov boundary as the Lorentz group on Minkowski space. Lorentz invariance is therefore a structural consequence of the D_5 cell geometry, not an external symmetry postulated on a Lagrangian. The Lorentzian signature is forced by the bounded-domain structure that ID1 establishes uniquely.

Significance: Completes the special-relativity correspondence from P1/P2/P3, answering the question Einstein left open in 1905: *why* spacetime is Lorentzian rather than Galilean or Euclidean. The signature $(3, 1)$ is not a choice but a theorem about D_5 geometry, and the Lorentz group is the isometry group of a structure forced by the cell-medium classification. Lorentz invariance has been verified to one part in 10^{20} in modern tests; QCT explains why it holds rather than treating it as an empirical input. (*ref:* Supp-1, § Lorentz.)

4 Electromagnetism

Source: QCT-Main §7, Supp-6.

4.1 EM1: Gauge Principle

Derives: The local $U(1)$ gauge structure of electromagnetism emerges from the phase freedom of cell-amplitude configurations along Hopf fibres.

Significance: Recovers the gauge principle as a structural consequence of cell-medium phase geometry, not as an external symmetry imposed on a Lagrangian. (*ref:* Main §7; Supp-6.)

4.2 EM2: Vector Potential as Hopf Connection

Derives: The electromagnetic vector potential A_μ is identified with the connection one-form on the Hopf bundle over the soliton moduli space.

Significance: Gives a geometric meaning to A_μ beyond its usual role as a gauge artefact; the four-potential becomes a curvature-bearing object on D_5 . (*ref:* Supp-6.)

4.3 EM3: Homogeneous Maxwell Equations

Derives: The Bianchi identities $\partial_{[\mu} F_{\nu\rho]} = 0$, equivalent to $\nabla \cdot B = 0$ and $\nabla \times E + \partial_t B = 0$, follow from $dF = 0$ for the Hopf curvature two-form $F = dA$.

Significance: Recovers Faraday's law of induction and the absence of magnetic monopoles as topological consequences of the cell connection. (*ref:* Supp-6.)

4.4 EM4: Lorentz Force Law

Derives: A charged soliton in a cell-medium gauge field experiences $F^\mu = qF^{\mu\nu}u_\nu$, derived from the soliton-photon vertex on D_5 .

Significance: Recovers the relativistic Lorentz force, the equation of motion underlying all of classical electromagnetism. (*ref:* Supp-6.)

4.5 EM5: Inhomogeneous Maxwell Equations

Derives: The source equations $\partial_\mu F^{\mu\nu} = J^\nu$, equivalent to Gauss's law and Ampère-Maxwell, follow from variation of the QCT gauge action with respect to A_μ .

Significance: Recovers Coulomb's law, current conservation, and electromagnetic radiation in a unified equation; QCT reproduces all four Maxwell equations from cell geometry. (*ref:* Supp-6.)

4.6 EM6: Photon Masslessness

Derives: The photon is the $D(4, [1, 0])$ representation in the $\text{Rac} \otimes \text{Rac}$ decomposition, which carries zero mass eigenvalue by the Casimir of the conformal group.

Significance: Photon masslessness becomes a structural consequence of D_5 representation theory rather than a separately imposed gauge-invariance condition. (*ref:* Supp-2/6.)

4.7 Electric Charge Quantisation from Hopf Topology

Derives: Electric charge in QCT is the Hopf winding number $Q \in \mathbb{Z}$, topologically protected by $\pi_3(S^2) = \mathbb{Z}$. Charge discreteness is therefore a theorem of sphere topology applied to the cell-amplitude soliton configuration, not an empirical observation requiring an external mechanism.

Significance: Replaces Dirac's monopole argument, which derives charge quantisation only at the cost of postulating an unobserved magnetic monopole, with a direct topological proof from the QCT soliton structure. Fractional charges are forbidden for free particles because no Hopf configuration carries non-integer winding, accounting for the universal absence of isolated fractional electric charges in all observations to date. (*ref:* Main paper §em1, charge quantisation.)

4.8 Coulomb's Law from $\Phi = c^2 \ln(s/s_0)$

Derives: The inverse-square electrostatic force law $F = kq_1q_2/r^2$ follows from linearising the QCT universal potential around a localised charge configuration.

Significance: Recovers the founding empirical law of electrostatics from the same potential that produces Newtonian gravity, demonstrating QCT's unification of static long-range forces. (*ref:* Main §7; Supp-6.)

5 Geometry and Cell Structure

Source: Supp-2.

5.1 ID1: $D_5 = \text{SO}_0(5, 2)/(\text{SO}(5) \times \text{SO}(2))$ Uniquely Forced

Derives: The BPS soliton moduli space is uniquely the irreducible bounded symmetric domain D_5 , established via parity involution, BPS Kähler geometry, Cartan classification, homogeneous-space extension, and a regularity closure ruling out proper subgroups of $\text{SO}_0(5, 2)$.

Significance: The central geometric identification of QCT: every subsequent structural result ($N_c = 20$, BPS saturation, the Wyler formula, the Yang-Mills coupling) rests on this domain being forced rather than chosen. (*ref:* Supp-2, § ID1.)

5.2 SC1: Dimension Five is Unique

Derives: The bounded symmetric domain dimension supporting the QCT soliton classification is uniquely $d = 5$; no other dimension admits a Cartan domain with the required Hopf/Joseph structure.

Significance: Answers why three spatial dimensions of physics emerge: D_5 has compatible boundary structure giving the observed $3 + 1$ Lorentzian signature; other Cartan domains do not. (*ref:* Supp-2, § SC1.)

5.3 $N_c = 20$ Cells per Soliton (Three Independent Proofs)

Derives: The ground-state $Q = 1$ Hopf soliton contains exactly twenty cells, established via three independent routes that converge on the same integer: (i) the Gelfand-Kirillov dimension of the completely prime Joseph ideal of $SO_0(5, 2)$ yields $\text{GKdim} = 4$, doubled by the Hopf S^1 fibre and refined by Rac/Di branching; (ii) the D_5 complex dimension 5 combined with the Joseph-ideal constraint forces $2 \times (5 + 5) = 20$; (iii) the Rac K-type decomposition with Hopf doubling counts twenty independent cell-amplitude modes in the ground state.

Significance: Fixes the fundamental cell count of physical particles as an exact structural theorem rather than a tunable parameter. Three independent proof routes converging on the same integer eliminate the possibility that $N_c = 20$ is a coincidence of one computation. Every subsequent QCT numerical prediction (the electron mass, the Yang-Mills coupling $g^2 = \alpha/14$, the fine-structure constant via Wyler, the CMB spectral tilt n_s , the proton-electron ratio) takes $N_c = 20$ as a fixed input, so the rigidity of this theorem propagates through the entire derivation chain. (*ref:* Supp-2, § Nc20.)

5.4 BPS Saturation: $E_2 = E_4$ Exactly

Derives: The $Q = 1$ Hopf soliton saturates the Bogomolny bound exactly, with gradient and curvature energies equal, for all configurations satisfying the Hopf constraint.

Significance: Provides the algebraic engine for the QCT mass-spectrum computations; without exact saturation, the proton-to-electron ratio and electron self-energy results would be approximate. (*ref:* Supp-2, § bps-saturation.)

5.5 Ground State Energy $E_0 = 3/2$ from Rac Casimir

Derives: The Rac singleton ground state energy is $E_0 = 3/2$, fixed by the D_5 Casimir condition $C_2(\text{Rac}) = E_0(E_0 - 4) = -15/4$.

Significance: Seeds the spectral tilt n_s , the coupling relation $\alpha = 4m_{\text{min}}^2/9$, and the BPS action; one of the most-reused numerical inputs in the derivation chain. (*ref:* Supp-2, § E0.)

5.6 Photon Emergence: $\text{Rac} \otimes \text{Rac}$ Contains $D(4, [1, 0])$ with Multiplicity 1

Derives: The Flato-Fronsdal decomposition $\text{Rac} \otimes \text{Rac} = \bigoplus_s D(s+3, [s, 0])$ contains the photon representation exactly once.

Significance: Explains why electromagnetism (rather than two or zero photons) emerges from soliton-soliton interaction; underpins the coupling derivation and gauge structure. (*ref:* Supp-2, § flato-fronsdal.)

6 Yang-Mills and the Mass Gap

Source: Supp-3.

6.1 Bridge Theorem: QCT D_5 IS $\text{SO}(5)$ Yang-Mills Theory

Derives: The continuum limit of the QCT D_5 lattice with Bergman measure is four-dimensional $\text{SO}(5)$ Yang-Mills theory in all gauge-invariant normalised expectation values: not merely equivalent to it via universality, but structurally identical, with the same link variable group element, the same measure up to a field-independent constant, and the same renormalisation group fixed point. Three independent facts establish this identity: (i) the link variable V_{ij} obtained from the Harish-Chandra section of $D_5 = G/K$ is exactly an $\text{SO}(5)$ group element, transforming under the standard Wilson gauge law; (ii) the Bergman measure on D_5 equals the Yang-Mills measure $[DA] \det(M_{\text{FP}})$ up to a field-independent constant $C(\varepsilon)$ that factors out of the path integral and cancels in every normalised correlator; (iii) the continuum limit flows to the same renormalisation group fixed point via Gross-Seiler (Proposition 5.3 of Supp-3) and Reisz (1988).

Significance: This is the formal grounding of QCT in real physics. Every QCT result that flows through the D_5 geometry is simultaneously a result about the actual physical universe, because $\text{SO}(5)$ Yang-Mills is a physical theory and QCT and $\text{SO}(5)$ Yang-Mills are the same theory in the gauge-invariant sector. QCT is not a model that resembles standard physics: it is standard physics, with the derivation running from three geometric postulates to the full structure rather than from observation to model. The Wyler derivation of $\alpha^{-1} = 137.036$ is therefore not a numerical coincidence but a theorem about the coupling constant of the theory that IS Yang-Mills in the continuum limit. Gap existence $\Delta > 0$, sufficient for the Clay requirement, follows from this bridge theorem alone. The numerical identification $\Delta = m_e$ is itself proved unconditionally within the QCT framework: the Rac soliton is the unique minimum-mass $Q = 1$ spin- $\frac{1}{2}$ excitation of QCT Yang-Mills theory, and all three defining invariants (topological charge $Q = 1$, spin $\frac{1}{2}$, minimum mass) are derived from P1, P2, P3 with no experimental input. The identification of the Rac soliton with the physical electron is therefore not a modelling assumption but the unique conclusion forced by P1/P2/P3, and so the Yang-Mills mass gap has a specific derived numerical value rather than merely a proof of existence. (*ref:* QCT-Prize-v31, § T5.1; QCT-Prize-Supp-v31, § continuum-limit; QCT-Supp-3-MassGap-v8, § 5.)

6.2 T5: Yang-Mills Existence and Mass Gap $\Delta = m_e$

Derives: A non-trivial Yang-Mills quantum field theory on $\mathbb{R}^4$ exists satisfying all five Wightman axioms, has strictly positive mass gap $\Delta_{\text{SO}(5)} = m_e > 0$ pinned to the electron

mass, and the construction extends to every compact simple Lie group G via embedding into $\text{SO}(N)$ for sufficiently large N . The proof assembles three components: T5.1 (Bergman-measure path integral with three-route coupling $g^2 = \alpha/14$), T5.2 (D_4 -lattice in the Wilson hypercubic universality class), and T5.3 (spectral gap survives the thermodynamic limit, pinned by the Vakulenko-Kapitanski topological bound).

Significance: This is the Clay Mathematics Institute Millennium Prize problem in Yang-Mills theory, unsolved since its 2000 formulation and one of the seven outstanding mathematical problems of the era. The QCT derivation produces both existence and a numerical gap value pinned to the measured electron mass, going beyond the prize statement (which asks only for a positive gap). The result rests on no postulate beyond P1, P2, P3 and structural theorems traced back to them. If sustained under formal review, it resolves the Yang-Mills problem from outside continuum QFT, identifying the mass gap with a topological invariant of D_5 rather than a non-perturbative dynamical scale. The numerical identification $\Delta = m_e$ is unconditional within the QCT framework: the three defining invariants of the Rac soliton (topological charge $Q = 1$, spin $\frac{1}{2}$, minimum mass) are each derived from P1, P2, P3 with no experimental input, so the identification of the Rac as the physical electron is forced, not assumed. (*ref:* Supp-3, § T5-complete.)

6.3 Wightman Axioms QCT-W1 through QCT-W5 Satisfied

Derives: All five Wightman axioms (relativistic covariance, vacuum uniqueness, spectrum condition, locality, and field domain) are satisfied by the QCT Hilbert space $\mathcal{H} = L^2(D_5, d\mu_{\text{Berg}})$, which is itself uniquely forced by the D_5 identification theorem ID1. Lorentz covariance follows from the Bergman isometry; the spectrum condition from BPS saturation and positivity of the Joseph-ideal Casimir; locality from the cell-medium support structure on D_5 .

Significance: Establishes QCT as a rigorous Wightman field theory in the sense of Streater-Wightman 1964, the mathematical scaffolding required for any Clay-grade Yang-Mills construction. No prior approach to the Yang-Mills problem has produced a full Wightman field theory satisfying all five axioms; constructive QFT has been able to do this only in low dimensions or for particular limits. The result is a prerequisite for the T5 mass-gap theorem, but is itself a structural achievement: it confirms that the QCT Hilbert space and its operator algebra meet the most demanding axiomatic standard available in mathematical physics. (*ref:* Supp-3, § wightman.)

6.4 Yang-Mills Coupling: $g^2 = \alpha/14$ (Three Routes Agree)

Derives: The Yang-Mills coupling satisfies $g^2 = \alpha/14$ for the D_5 geometry, computed independently via Harish-Chandra integration, Wilson lattice BCH expansion, and Killing-form index contraction.

Significance: Pins the strong-sector coupling to the fine-structure constant by a numerical identity; the consistency of three derivation routes is a non-trivial cross-check. (*ref:* Supp-3, § T5.1.)

6.5 No Massless Goldstone Bosons

Derives: The $Q = 0$ sector of QCT contains no massless Goldstone modes; the global symmetry structure of D_5 does not produce them.

Significance: Removes a potential infrared obstruction to the mass gap; the gap is not evaded by pseudo-Goldstone modes. (*ref:* Supp-3, § goldstone.)

6.6 Partition Function Normalisation: $Z[0] = 1$

Derives: The QCT path integral partition function satisfies $Z[0] = 1$ in every topological Q sector simultaneously.

Significance: A normalisation prerequisite for the Wightman verification and the mass-gap proof; without this, expectation values would be ill-defined. (*ref:* Supp-3, § C11-Z0.)

6.7 Coupling Relation $\alpha = 4m_{\min}^2/9$

Derives: The fine structure constant and the minimum soliton mass satisfy $\alpha = 4m_{\min}^2/9$, derived from $N_c = 20$ and $E_0 = 3/2$.

Significance: Links α and m_e as mutually determined geometric invariants; removes the apparent independence of two of the most-measured constants in physics. (*ref:* Supp-3, § alpha-constant.)

6.8 T5.2: D_4 Lattice in the Wilson Universality Class

Derives: The QCT D_4 -lattice lies in the Wilson hypercubic universality class, with convergent polymer expansion, ellipticity constant $c_{D_4} = 6 > c_{\text{hyp}} = 2$, and $\nabla R = 0$ eliminating field-dependent one-loop corrections.

Significance: Places the QCT path-integral measure in the correct universality class for the Yang-Mills continuum limit; a Clay-grade requirement. (*ref:* Supp-3, § T5.2.)

6.9 T5.3: Spectral Gap Survives the Thermodynamic Limit

Derives: The Yang-Mills lower bound $\Delta_\infty \geq m_e$ is preserved as $N \rightarrow \infty$, established via transfer-matrix methods, Mosco convergence, and the Vakulenko-Kapitanski topological bound.

Significance: The non-trivial step in the Clay problem: gaps in finite-volume lattice theories typically close as the volume grows; in QCT the gap is pinned by topology. (*ref:* Supp-3, § T5.3.)

6.10 Bergman Measure Equals Faddeev-Popov Determinant

Derives: The Bergman measure on D_5 coincides exactly with the Faddeev-Popov gauge-fixing determinant, $J_{\text{Berg}} = C \cdot \det(M_{\text{FP}})$.

Significance: Justifies the Bergman path integral as a properly gauge-fixed Yang-Mills path integral; closes a structural prerequisite for T5. (*ref:* Supp-3, § J-berg.)

7 Particle Masses

Source: Supp-4.

7.1 Proton-to-Electron Mass Ratio: $m_p/m_e = 6\pi^5$ (Tree Level)

Derives: At tree level $m_p/m_e = 6\pi^5 \approx 1836.12$, computed from five independent geometric factors of the compact dual Q_5 coset space (volume ratios, Kähler cohomology, normalised Bergman measure, and BPS saturation factors). No mass parameters, no Yukawa couplings, and no quark masses enter the derivation; the result is fixed by D_5 geometry alone. The two-loop correction with coefficient $9/(8\pi)$ extends the precision further.

Significance: Predicts the CODATA value 1836.15267343(11) to four significant figures from pure geometry, with the next correction structurally fixed. This is among the most striking unforced numerical predictions in the QCT programme: in the Standard Model the proton mass is a non-perturbative QCD quantity with no closed-form expression, and the proton-electron ratio is a contingent fact requiring lattice QCD with quark-mass inputs. QCT produces it as a closed-form geometric identity at tree level, with the geometric origin of the factor π^5 identified with the Kähler cohomology of Q_5 . (*ref:* Supp-4, § proton-GK.)

7.2 One-Loop Electron Self-Energy Coefficient: $C = 3\alpha/(8\pi)$

Derives: The one-loop electron self-energy coefficient in QCT-QED is $C = 3\alpha/(8\pi)$, derived from the QCT Lagrangian without importing external QED renormalisation conventions.

Significance: Pins the leading radiative correction to the electron mass from first principles; first step toward sub-ppm electron mass prediction. (*ref:* Supp-4, § one-loop.)

7.3 Two-Loop Electron Mass Coefficient: $C_2^{\text{QCT}} = -0.13016$

Derives: The two-loop correction to the QCT electron self-energy has coefficient $C_2^{\text{QCT}} = -0.13016$.

Significance: Together with the one-loop result, closes the gap between tree-level and CODATA electron mass to sub-ppm precision. (*ref:* Supp-4, § two-loop.)

7.4 All-Loop QED Halving: $\text{QCT} = \frac{1}{2} \times \text{QED}$

Derives: The QCT n -loop self-energy coefficient is exactly half the corresponding QED value at every loop order $n \geq 1$.

Significance: A universal structural relation between QCT and QED radiative corrections; allows all-order mass computations to inherit known QED loop integrals. (*ref:* Supp-4, § all-loop-halving.)

7.5 Muon Mass Formula

Derives: The muon-to-electron mass ratio follows from the QCT excited-soliton spectrum within the same D_5 representation framework that produces m_p/m_e .

Significance: Extends the QCT mass-spectrum chain from the proton-electron pair to the second-generation lepton; brings the muon mass into the same geometric origin. (*ref:* Supp-4.)

8 Cosmology

Source: QCT-Main, Supp-5.

8.1 CMB Spectral Tilt: $n_s = 1 - 5e^{-\pi^2/2}$

Derives: The primordial scalar spectral index is $n_s = 1 - 5e^{-\pi^2/2} \approx 0.9647$, derived from the leading non-trivial eigenvalue of the D_5 heat-kernel acting on Phase 1 cell-amplitude perturbations. The factor of 5 counts the relevant Rac modes; the exponent $\pi^2/2$ traces to the D_5 Casimir gap $E_0 = 3/2$ on the Shilov boundary. No inflaton potential, no slow-roll parameters, and no free tuning enter.

Significance: Predicts the Planck 2018 measurement $n_s = 0.9649 \pm 0.0042$ to better than 0.02%, an agreement that no inflationary model attains without parameter fitting. The result demonstrates that the CMB spectral tilt is a structural invariant of D_5 rather than a free inflationary input, recasting one of the central observables of modern cosmology as a D_5 theorem. Combined with n_s , the predicted tensor-to-scalar ratio and running provide falsifiable signatures for CMB-S4 and LiteBIRD. (*ref:* Supp-5, § ns-formula.)

8.2 Early-Universe Equation of State: $w = 1/3$

Derives: The Phase 1 equation of state $w = p/\rho = 1/3$ follows from P1 (energy conservation) and P2 (scale-free coupling) without thermodynamic input.

Significance: Recovers the radiation-dominated background of standard cosmology from first principles; the dynamical setting for the n_s and baryogenesis calculations. (*ref:* Supp-5, § w13.)

8.3 D_5 Geometry Present in Phase 1 (Pre-Soliton Era)

Derives: The D_5 structure is a property of the cell medium itself in Phase 1, prior to any soliton formation.

Significance: Validates the use of D_5 spectral data for cosmological initial conditions; CMB observables can be computed before particle physics is in place. (*ref:* Supp-5, § phase1-D5.)

8.4 Baryon Asymmetry: $\eta = 6.18 \times 10^{-10}$

Derives: The baryon-to-photon ratio is fixed by a parity-asymmetric topological count of Hopf-charge configurations in the Phase 1→Phase 2 cell-medium transition that produced the first solitons, giving $\eta \approx 6.18 \times 10^{-10}$. The asymmetry arises from the parity involution on D_5 that distinguishes $Q = +1$ from $Q = -1$ solitons at the transition; no additional CP-violating sector, leptogenesis mechanism, or sphaleron process is invoked.

Significance: Matches the BBN/CMB-inferred value $\eta = (6.12 \pm 0.04) \times 10^{-10}$ from first principles, addressing one of the longest-standing open problems in cosmology (why there is matter rather than equal matter and antimatter) without invoking new physics beyond P1/P2/P3. Sakharov's three conditions (baryon-number violation, CP violation, departure from equilibrium) are all satisfied structurally by the cell-medium phase transition: baryon-number violation comes from topological charge non-conservation across the transition,

CP violation from the D_5 parity involution, and departure from equilibrium is automatic for a first-order phase transition. (*ref*: Supp-5, § baryogenesis.)

8.5 MOND Acceleration Scale a_0 from $\Phi = c^2 \ln(s/s_0)$

Derives: The crossover acceleration between Newtonian and modified-dynamics regimes is fixed by the logarithmic universal potential, yielding the MOND scale $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$.

Significance: Recovers Milgrom’s empirical scale from the same potential that gives Newtonian gravity; explains flat galactic rotation curves without dark matter as a separate substance. (*ref*: Supp-5.)

8.6 Flat Galactic Rotation Curves

Derives: The asymptotic flatness of galactic rotation velocities follows from the logarithmic regime of $\Phi = c^2 \ln(s/s_0)$ at low cell-density gradients.

Significance: Reproduces the central observation that motivated dark matter and MOND alike; QCT delivers the flat curves from cell-medium dynamics directly. (*ref*: Supp-5.)

8.7 Baryonic Tully-Fisher Relation

Derives: The observed $v_{\text{flat}}^4 \propto M_{\text{baryonic}}$ relation follows from the logarithmic potential combined with cell-count conservation.

Significance: Recovers a tight empirical scaling law that is unexplained in Λ CDM but is structural in QCT. (*ref*: Supp-5.)

8.8 Dark Matter as Cell Compression

Derives: The phenomenology attributed to dark matter (flat rotation curves, weak lensing in excess of baryonic mass, cluster dynamics, the Bullet Cluster offset) is reproduced by cell-density compression around baryonic structures under P2, with no second particle species required. The cell medium itself compresses in response to baryonic mass, producing additional gravitational pull that scales with the baryonic distribution but is offset from it during dynamical events. The Bullet Cluster offset emerges naturally because cell compression follows the dominant gravitational potential, not the visible gas.

Significance: Provides a single-substance account of galactic dynamics, lensing, and cluster mass profiles within QCT, addressing the most thoroughly tested gap between Λ CDM and observation. Unlike pure MOND, the QCT mechanism handles both galactic rotation curves and cluster-scale lensing within the same framework. The account is testable against detailed lensing maps, dwarf-galaxy dynamics, and the small-scale structure puzzles (missing satellites, too-big-to-fail, core-cusp) where particle dark matter is under tension. (*ref*: Supp-5.)

8.9 Logarithmic Spirals as Dynamical Attractors

Derives: Cell-medium fluid mechanics drives spiral arms toward logarithmic spiral geometry as a dynamical attractor. The spiral structure is a stable solution of the cell-flow equation

rather than a transient material pattern requiring ongoing maintenance, so differential rotation does not wind it up.

Significance: Resolves the long-standing winding problem, the puzzle of why spiral arms do not wind up and disappear over billions of years of differential rotation, by replacing the vacuum assumption of standard astrophysics with a fluid medium in which spiral geometry is naturally self-sustaining. Density wave theory, magnetic braking, and dark-matter halo stabilisation are all rendered unnecessary. (*ref:* Main paper §P20, sec:spiral-gap.)

8.10 Halo Truncation Radius $R_{\text{halo}} = v_{\text{rot}}^2/a_0$

Derives: The effective galactic dark-matter halo has a sharp truncation radius $R_{\text{halo}} = v_{\text{rot}}^2/a_0$ beyond which cell compression from the host galaxy becomes sub-dominant to the cosmological background. The predicted value $R_{\text{halo}} \approx 14.5 \text{ kpc}$ for the Milky Way is consistent with the pitch-angle constraint from spiral arm geometry.

Significance: Predicts a specific, measurable boundary to the effective dark-matter halo. The prediction is consistent with Gaia DR4 stellar kinematics and is structurally distinct from the diffuse NFW profiles of ΛCDM , which have no analogous sharp truncation scale. (*ref:* Main paper §predictions.)

8.11 Ring Galaxy Inner Rings from Phase Boundary Reflection

Derives: Phase boundary reflection in the cell medium produces secondary inner rings in ring galaxies, with the ring geometry determined by the cell-compression wavefront dynamics following a galactic collision.

Significance: Predicts the confirmed inner-ring structure of the Cartwheel galaxy from first principles. Standard collisional ring formation does not generically predict inner rings, making this a discriminating observational test between QCT and collisionless ring models. (*ref:* Main paper §predictions.)

8.12 Bar Formation from Cell Medium Viscosity

Derives: Whether a galactic bar instability develops is determined by the viscosity of the cell medium relative to the disk-to-halo mass ratio. Bar formation is a viscosity-threshold phenomenon in the QCT fluid rather than a purely gravitational instability.

Significance: Provides a first-principles mechanism for galactic bar formation and predicts a correlation between bar fraction and cell-medium viscosity that is testable against the morphological distribution of galaxies in large surveys such as SDSS and Euclid. (*ref:* Main paper §predictions.)

8.13 MOND Acceleration Scale Evolves with Redshift: $a_0(z) = cH(z)/2\pi$

Derives: The MOND transition acceleration is not a universal constant but tracks the Hubble rate at each cosmic epoch, $a_0(z) = cH(z)/2\pi$. Higher-redshift galaxies therefore form with tighter spiral pitch angles, lower bar fractions, and smaller effective haloes, since each of those structural features is set by the local value of a_0 at formation.

Significance: A sharp, falsifiable prediction distinguishing QCT from both Λ CDM, which has no a_0 , and standard MOND, which treats a_0 as a universal constant. The predicted redshift dependence of galaxy morphology is directly testable with JWST and Euclid rotation curves at $z > 1$, where any of the three signatures (tighter spirals, fewer bars, smaller haloes) would corroborate the QCT-predicted scaling, and the absence of all three would falsify it. (*ref:* Main paper §force-mond; QCT-Supp-5 C1.)

8.14 Bullet Cluster: Lensing Tracks Baryons, Not Collisionless Dark Matter

Derives: The cell medium responds to the gravitational potential of baryonic mass and produces gravitational lensing that tracks the galaxy concentrations of the post-collision Bullet Cluster rather than a collisionless dark-matter component. The observed lensing-to-baryon offset is reproduced from cell-medium compression dynamics alone, with no exotic particle species invoked.

Significance: The Bullet Cluster is standardly cited as the strongest observational argument for particle dark matter, since its lensing peaks are offset from its X-ray gas. QCT reproduces the observation as a structural consequence of cell-medium response to galactic potentials, removing what was considered a near-decisive case for dark matter as a separate substance. (*ref:* Main paper §nodark.)

8.15 Cosmological Constant: $\Lambda(\alpha)$

Derives: The apparent vacuum energy density is fixed by a function of α alone, $\Lambda = \Lambda(\alpha)$, with no separately tunable parameter. The specific form, $\rho_\Lambda/\rho_P = e^{-2/\alpha}$ (made precise in the Dark Matter and Dark Energy section), pins the cosmological-constant scale to the fine-structure constant via an exponential suppression.

Significance: Addresses the cosmological constant problem, the worst quantitative disagreement between theory and observation in physics (traditionally requiring 120 orders of magnitude of fine-tuning between bare and renormalised vacuum energy), by pinning Λ to α via a structural identity. With $\alpha \approx 1/137$, $e^{-2\alpha^{-1}} \approx e^{-274}$, an exponential factor that naturally produces the observed $\rho_\Lambda/\rho_P \sim 10^{-120}$ suppression without tuning. (*ref:* Supp-5.)

8.16 Hubble Constant: $H_0 = 69.6 \text{ km/s/Mpc}$

Derives: The present-day expansion rate evaluated within the QCT cosmological model is $H_0 = 69.6 \text{ km/s/Mpc}$.

Significance: Lands between the Planck CMB value (67.4) and the SH0ES local value (73.0); offers a calculation-based path through the Hubble tension rather than a new parameter. (*ref:* Supp-5.)

8.17 Hubble Tension Resolution Mechanism

Derives: The early-universe inference channel (CMB acoustic scale at recombination, dominated by D_5 -phase physics) and the late-universe channel (distance ladder via Cepheids and Type Ia supernovae, dominated by soliton-phase physics) measure structurally different

effective expansion rates in QCT, with a calculable offset arising from the Phase 1→Phase 2 cell-medium transition. The CMB channel returns the Phase 1 effective H_0 , the distance-ladder channel returns the present-day soliton-phase value, and the gap between them is set by the cell-compression history.

Significance: Explains the 5σ Hubble tension, the discrepancy between Planck CMB (67.4 ± 0.5 km/s/Mpc) and SH0ES distance-ladder (73.0 ± 1.0 km/s/Mpc) determinations of H_0 , currently the most prominent crisis in observational cosmology, without invoking early dark energy, modified recombination physics, or new neutrino species. The two measurements are not in conflict; they are measuring different quantities that QCT predicts to differ. (*ref:* Supp-5.)

8.18 Impossibly Early Galaxy Problem Resolved

Derives: Earlier and more massive galaxy formation than standard Λ CDM predicts follows from cell-compression dynamics that boost effective gravitational pull at high redshift.

Significance: Accommodates JWST observations of mature galaxies at $z \gtrsim 10$, which strain standard cosmology, as a structural prediction of QCT. (*ref:* Supp-5.)

8.19 Arrow of Time from Cell-Count Conservation

Derives: The thermodynamic arrow of time emerges from the monotonic growth of accessible cell configurations under P1 conservation and P2 resizing.

Significance: Provides a structural explanation for time’s directionality grounded in cell-medium dynamics rather than low-entropy initial conditions alone. (*ref:* Supp-5.)

8.20 Black-Hole Information Conservation

Derives: Information is preserved across QCT sonic horizons because cell count is conserved globally under P1; no information is destroyed at horizon crossing.

Significance: Resolves the black-hole information paradox in QCT’s favour without invoking firewalls or non-unitary evolution. (*ref:* Supp-5.)

9 Force Unification and Predictions

Source: Supp-6.

9.1 Master Force Equation (Gravity, EM, Nuclear in One Form)

Derives: A single force law derived from the universal potential $\Phi = c^2 \ln(s/s_0)$ reduces, in different cell-density regimes, to Newtonian gravity ($\rho \rightarrow 0$ limit), Coulomb’s law (intermediate-density localised charges), the MOND crossover (low-acceleration regime $a < a_0$), and nuclear-scale binding (cell-compression regime near $\rho \rightarrow \rho_P$). Each regime corresponds to a different linearisation of the same logarithmic potential.

Significance: Demonstrates structural unification of the long-range forces and the nuclear sector within QCT, going beyond the partial unification achieved by GUT and string approaches (which unify gauge sectors but not gravity). The regime-by-regime reduction

makes the unification claim operational and testable: each force is a calculable limit of the master equation, and the crossover scales between regimes are themselves predictions. This is a stronger structural claim than coupling-constant unification: it identifies all forces as one cell-medium dynamics seen at different densities. (*ref*: Supp-6.)

9.2 MOND Crossover from the Master Equation

Derives: The transition between Newtonian and MOND-like regimes occurs at a calculable acceleration scale built into the universal potential’s logarithmic structure.

Significance: Recovers Milgrom’s interpolation as a derived feature rather than an empirical fit, with the crossover scale fixed by QCT geometry. (*ref*: Supp-6.)

9.3 Force Unification at Soliton Scale

Derives: At the soliton scale, gravitational, electromagnetic, and short-range cell-medium interactions become structurally equivalent expressions of the same cell-density dynamics.

Significance: Provides QCT’s substitute for a grand unification scale, with the unification realised by cell-medium geometry rather than gauge-group embedding. (*ref*: Supp-6.)

9.4 Weinberg Angle from D_5 Representation Theory

Derives: The electroweak mixing angle $\sin^2 \theta_W$ is computed from the branching of the relevant D_5 representations under the electroweak subgroup.

Significance: Reproduces a Standard Model parameter measured to high precision (at M_Z , $\sin^2 \theta_W \approx 0.231$) from QCT geometry, with no tunable input. (*ref*: Supp-6.)

9.5 Fine Structure Constant: Wyler Formula from P1/P2/P3

Derives: The Wyler formula $\alpha = (9/8\pi^4)(\pi^5/1920)^{1/4}$ is derived in full from P1/P2/P3, with all three historical gaps in the chain closed: (i) the choice of D_5 as the relevant domain (closed by ID1’s uniqueness theorem); (ii) the Bergman measure normalisation (closed by the $J_{\text{Berg}} = C \cdot \det(M_{\text{FP}})$ identity); and (iii) the G -equivariance of the soliton-photon coupling map (closed in Supp-6, forcing $\alpha = \alpha_{\text{Wyler}}$).

Significance: Predicts $\alpha^{-1} \approx 137.036$ from pure geometry, within 0.61 ppm of the CODATA value 137.035999084(21), and turns Wyler’s 1969 numerological coincidence into a structural theorem. The fine structure constant is among the most thoroughly measured numbers in physics; explaining its specific value from first principles has been an open challenge for a century, asked by Eddington, Feynman (“one of the greatest damn mysteries of physics”), and Pauli. QCT identifies α as the ratio of two computable volumes on D_5 , structurally fixed by the same geometry that produces $N_c = 20$ and the Yang-Mills coupling. (*ref*: Supp-2/6, §wyler.)

9.6 G -Equivariance and $\alpha = \alpha_{\text{Wyler}}$

Derives: The soliton-photon coupling map is G -equivariant, which forces $\alpha = \alpha_{\text{Wyler}}$ as a structural consequence of the QCT measure.

Significance: Closes the final logical gap in the fine-structure-constant derivation; equivariance promotes the Wyler value from numerical match to derived identity. (*ref:* Supp-2/6, § wyler-equivariance.)

9.7 Unified Potential $\Phi = c^2 \ln(s/s_0)$ Operative Across Domains

Derives: The scalar field $\Phi = c^2 \ln(s/s_0)$ functions as QCT’s universal potential, with the same expression governing gravity, electromagnetism, quantum mechanics, and cosmology.

Significance: Removes the need for separate potential ansätze in each domain; the same logarithmic structure produces Newtonian gravity, Coulomb’s law, and the MOND scale. (*ref:* Supp-1, § universal-field; Supp-6.)

10 Chemistry

Source: Supp-7.

10.1 CH1: Periodic Table Bounded by α Alone

Derives: The maximum stable atomic number is fixed by the fine structure constant via the QCT cell-compression bound $Z_{\max} \lesssim 1/\alpha$, with no nuclear input beyond α and the cell geometry.

Significance: Recovers the empirical observation that the periodic table ends near $Z \sim 118$ as a structural consequence of α , rather than a contingent fact of nuclear stability. (*ref:* Supp-7, § sec:supp-ch1.)

10.2 CH2: Extended Periodic Table at Reduced α

Derives: If α were reduced, the bound $Z_{\max} \sim 1/\alpha$ shifts upward, predicting the location of the hypothetical extended periodic table and the next noble-gas closures.

Significance: Turns “what if the constants were different?” into a falsifiable structural prediction about chemistry in alternative cell-medium parameter regimes. (*ref:* Supp-7, § sec:supp-ch2.)

10.3 CH3: Chemical Bonds from Cell Compression

Derives: Covalent and ionic bond energies follow from the cell-compression cost of bringing two atomic cell-clouds into overlap, computed from the same logarithmic potential that governs gravity and electrostatics.

Significance: Places chemistry on the QCT derivation chain: bond energies are not a separate empirical layer but a cell-medium consequence of the universal potential. (*ref:* Supp-7, § sec:supp-ch3.)

10.4 CH6: Dirac Exchange Functional from P2

Derives: The Kohn–Sham local-density exchange functional $E_x^{\text{LDA}}[\rho] = -C_x \int \rho^{4/3} d^3r$ with the standard coefficient $C_x = (3/4)(3/\pi)^{1/3}$ follows from cell-resizing under P2 applied to a uniform electron gas.

Significance: Recovers the workhorse exchange functional of density-functional theory, the basis of essentially all modern computational chemistry, from a single QCT postulate. (*ref:* Supp-7, § sec:supp-ch6.)

10.5 CH7: Spin-Valley Locking as Hopf Winding Conservation

Derives: Spin-valley locking in two-dimensional materials with broken inversion symmetry is identified with conservation of the Hopf winding number on the QCT soliton moduli space; the coupling between spin and valley degrees of freedom is a topological invariant, not an emergent band-structure accident.

Significance: Provides a structural, first-principles origin for a class of condensed-matter selection rules currently treated as material-specific band-structure features; the topological protection predicts the absence of spin-valley relaxation channels across the entire material class, with broad implications for valleytronics and spintronic device design. (*ref:* Supp-7, § sec:supp-ch7.)

11 Extended Particle Masses

Source: Supp-8.

11.1 Proton Tree-Level Mass: $m_p/m_e = 6\pi^5$ (Five-Factor Proof)

Derives: The proton-to-electron mass ratio at tree level is $6\pi^5$, established by a five-factor decomposition over the compact dual Q_5 with cancellation of path-integral normalisations and a Kähler-cohomology factor of π^5 .

Significance: Promotes the m_p/m_e derivation from a single coset computation to a five-independent-factors structural theorem; CODATA agreement to four significant figures with no tunable parameters. (*ref:* Supp-8, § sec:supp-proton-gapA.)

11.2 Proton Two-Loop Radiative Coefficient $9/(8\pi)$

Derives: The two-loop QCT radiative correction coefficient for the proton mass is $9/(8\pi)$, derived from the QCT-QED loop integrals on Q_5 without external renormalisation input.

Significance: Closes the radiative-correction chain for the proton mass and pins the leading deviation from $6\pi^5$ to a calculable rational/ π coefficient. (*ref:* Supp-8, § sec:supp-proton-alpha2.)

11.3 Muon as the Di Singleton on $\check{S}(D_5)$

Derives: The muon is identified with the Di singleton representation of $\text{SO}_0(5, 2)$ on the Shilov boundary $\check{S}(D_5) = (S^4 \times S^1)/\mathbb{Z}_2$, distinct from the electron’s bulk Rac representation.

Significance: Gives the muon a structural identity in the same D_5 framework that fixes the electron and proton; second-generation leptons become boundary excitations rather than scaled copies. (*ref:* Supp-8, §sec:supp-muon-di.)

11.4 Tanaka–Webster Invariant of $\check{S}(D_5)$

Derives: The Tanaka–Webster CR invariant of the Shilov boundary $\check{S}(D_5)$ is computed in closed form and used as the geometric input to the muon tree-level mass.

Significance: Identifies a previously-unrecognised CR-geometric invariant as the load-bearing quantity for second-generation lepton masses. (*ref:* Supp-8, §sec:supp-muon-tw.)

11.5 Muon Tree-Level Mass from Rumin Volume

Derives: The tree-level muon-to-electron mass ratio follows from the Rumin volume of $\check{S}(D_5)$ normalised against the bulk D_5 Bergman measure.

Significance: Reproduces m_μ/m_e at tree level from pure geometry; closes the leading geometric piece of the lepton-mass hierarchy. (*ref:* Supp-8, §sec:supp-muon-treelevel.)

11.6 Root-Channel Theorem and One-Loop Muon Mass

Derives: The one-loop QED correction to the muon mass decomposes along the B_2 root channels of D_5 , with the photon-overlap channels carrying the dominant contribution.

Significance: Extends the all-loop halving structure of QCT-QED to the muon, producing the full one-loop muon mass formula with sub-percent agreement to the measured value. (*ref:* Supp-8, §§sec:supp-muon-oneloop, sec:supp-muon-complete.)

11.7 Photon-Overlap Equality and the 1/3 Ratio

Derives: The photon-overlap integrals of the three muon one-loop channels are exactly equal, and the long/short-channel ratio is exactly 1/3 at one loop.

Significance: An unforced rational number emerging at one loop; a structural cross-check that the D_5 geometry is doing real work, not absorbing free parameters. (*ref:* Supp-8, §sec:supp-muon-overlap.)

11.8 Koide Algebraic Equivalences

Derives: The classical Koide identity $K = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ is shown equivalent to a family of algebraic statements about the QCT lepton mass triple on B_2 .

Significance: Reframes a long-standing empirical regularity of the charged-lepton spectrum as a derivable structural identity within QCT. (*ref:* Supp-8, § Koide algebraic equivalences.)

11.9 Arithmetic-Mean-of-Extremes Identity

Derives: The QCT lepton-mass triple satisfies an arithmetic-mean-of-extremes identity that pins the middle (muon) mass between the electron and tau as the algebraic mean of the constraint-surface extrema.

Significance: A new exact algebraic identity in the lepton sector, derived in QCT and consistent with measurement; testable by future precision determinations of m_τ . (ref: Supp-8, § Arithmetic-mean-of-extremes.)

11.10 Tau Mass from Koide Closure

Derives: The tau-lepton mass follows from closing the Koide relation on the corrected electron and muon masses derived earlier in Supp-8.

Significance: Closes the charged-lepton hierarchy: m_e, m_μ, m_τ are all fixed within QCT from D_5 geometry and the Koide identity, with no third lepton mass introduced as free input. (ref: Supp-8, § sec:supp-tau.)

11.11 B_2 Restricted-Root Origin of the 1/3/9 Pattern

Derives: The 1 : 3 : 9 multiplicity pattern observed in the QCT radiative-correction coefficients is shown to follow from the restricted-root multiplicities of the B_2 subsystem of D_5 .

Significance: Replaces a numerical pattern in the loop calculations with a representation-theoretic theorem; the 1/3/9 ratios become forced by D_5 Lie theory. (ref: Supp-8, § sec:supp-B2-pattern.)

12 Nuclear Physics

Source: Supp-9.

12.1 Topological Confinement of Fractional Charge

Derives: The QCT Hopf charge Q commutes with the Hamiltonian and is integer-valued, so configurations with non-integer Q are superselected out; in particular, isolated quark-like fractional charges cannot appear as asymptotic states.

Significance: Recovers quark confinement as a topological theorem rather than a dynamical conjecture; relocates the confinement problem from non-perturbative QCD into D_5 Hopf topology. (ref: Supp-9, § sec:strong-force.)

12.2 Charge Quantisation Forbids Free Quarks

Derives: Since electric charge in QCT is the integer Hopf winding number $Q \in \mathbb{Z}$ and no topological configuration on S^3 carries $Q = 1/3$ or $Q = 2/3$, free quarks are topologically forbidden as asymptotic states. The quark degrees of freedom appearing in deep-inelastic scattering are collective modes of multi-soliton configurations rather than detachable fractional-charge particles.

Significance: Provides a topological explanation for quark confinement that is independent of the strong-coupling non-perturbative QCD argument. The complete absence of isolated fractional electric charges across every experimental search to date follows directly, rather than as a phenomenological observation awaiting a dynamical explanation. (*ref:* Main paper §predictions; Supp-9, §sec:strong-force.)

12.3 Strong Force as Cell-Medium Confinement

Derives: The short-range nuclear binding force is identified with the cell-compression interaction between Hopf solitons of charge $Q = 1$, with strength set by the QCT coupling $g_0 = \sqrt{\alpha}$.

Significance: Provides a structural account of the strong force without a separately postulated gauge sector; the same cell-medium dynamics that produce gravity and EM also produce hadronic binding. (*ref:* Supp-9, §sec:strong-force.)

12.4 Asymptotic Freedom from $g(\rho)$

Derives: The cell-medium coupling $g(\rho)$ flows from $g_0 = \sqrt{\alpha}$ at low density to $g \rightarrow 1$ as $\rho \rightarrow \rho_P$, reproducing asymptotic freedom in the QCT strong sector.

Significance: Recovers the running of the strong coupling, the empirical hallmark of QCD, from cell-density dynamics rather than perturbative beta-function input. (*ref:* Supp-9, §sec:strong-force, asymptotic freedom.)

12.5 Semi-Empirical Mass Formula Terms from Cell Dynamics

Derives: The volume, surface, Coulomb, asymmetry, and pairing terms of the nuclear semi-empirical mass formula are each identified with a specific cell-medium mechanism (P1 compression, boundary deficit, P2 long-range, isospin counting, soliton-pair binding).

Significance: Places nuclear binding energetics on the QCT derivation chain; the Weizsäcker formula becomes a structural decomposition rather than an empirical fit. (*ref:* Supp-9, §sec:binding.)

13 Neutrinos

Source: Supp-10.

13.1 Three-Stratum Mass Architecture on D_5

Derives: Lepton masses decompose across three geometric strata of D_5 : the bulk (electron, Rac singleton), the Shilov boundary $\check{S}(D_5)$ (muon, Di singleton), and the deeper boundary stratum supporting neutrinos.

Significance: Provides a structural reason for three lepton generations grounded in D_5 stratification; generations are not copies but geometrically distinct boundary types. (*ref:* Supp-10, §sec:strata.)

13.2 Muon-Electron Mass Ratio from Shilov Boundary

Derives: The muon-to-electron mass ratio is fixed by the Shilov-boundary geometry of D_5 , with the explicit boundary-to-bulk normalisation reproducing the measured ratio to within radiative corrections.

Significance: A second-generation lepton mass derived from boundary geometry rather than introduced as a free Yukawa coupling; reused in the Koide closure for m_τ . (*ref:* Supp-10, § sec:strata, muon mass.)

13.3 Koide $K = 2/3$ from B_2 Geometry

Derives: The Koide ratio $K = 2/3$ is derived from the B_2 restricted-root structure of D_5 , independently of the algebraic-extremality argument in Supp-8.

Significance: A second, geometry-side derivation of the Koide identity; the agreement of the algebraic and geometric routes is a non-trivial cross-check of the lepton-sector framework. (*ref:* Supp-10, § sec:koide.)

13.4 GKdim Algebraic Gap for $SO_0(5, 2)$

Derives: The Gelfand–Kirillov dimension of the relevant $SO_0(5, 2)$ ideal admits a strictly positive algebraic gap, established as Confirmation A in the QCT prize supplement.

Significance: A structural algebraic gap that constrains the neutrino-sector representation theory and is reused in the spin-statistics theorem for finite N . (*ref:* Supp-10, § sec:neutrinos, GKdim=4.)

13.5 PMNS Matrix Structure from D_5 Symmetry Breaking

Derives: The neutrino mixing (PMNS) matrix structure is derived from the symmetry-breaking pattern of D_5 across the three lepton strata, predicting the observed near-tribimaximal hierarchy.

Significance: Places the PMNS matrix on the QCT derivation chain; flavour mixing becomes a geometric stratification statement rather than an independent flavour sector. (*ref:* Supp-10, § sec:neutrinos, PMNS.)

14 Quantum Gravity

Source: Supp-11.

14.1 Cell-Size Field as the Graviton

Derives: The cell-size field $s(x)$ of P2 is identified as the QCT graviton, a single scalar degree of freedom whose excitations carry effective spin-2 character through the cell-tensor structure.

Significance: Recovers gravity as a quantum field without introducing a separate metric or spin-2 field; the same field that resizes cells under P2 is the graviton. (*ref:* Supp-11, § sec:graviton.)

14.2 Shift Symmetry of the Cell-Size Field

Derives: The cell-size field admits an exact shift symmetry $s \rightarrow s + c$ under P2 invariance, with the symmetry preserved by the QCT action.

Significance: Forces the graviton to be exactly massless and gravity to be exactly long-range as corollaries; closes the massless-graviton question structurally. (*ref:* Supp-11, §sec:graviton, shift symmetry.)

14.3 Graviton Exactly Massless; Gravity Long-Range

Derives: As corollaries of the shift symmetry, the QCT graviton has exactly zero mass and gravity is a strictly long-range force with no Compton-wavelength cutoff.

Significance: Removes the need to fine-tune the graviton mass to zero; matches the empirical bounds on graviton mass from binary pulsar timing and LIGO observations. (*ref:* Supp-11, §sec:graviton.)

14.4 Gravitational Cell-Size Profile (Standalone Theorem)

Derives: The static cell-size profile around a localised mass solves a definite QCT field equation whose solution reproduces, in the weak-field limit, the Schwarzschild spatial profile to leading post-Newtonian order.

Significance: Provides a standalone derivation of the gravitational profile from cell dynamics; the input to the GR1–GR7 tests catalogued earlier. (*ref:* Supp-11, §sec:grav-profile.)

14.5 Universal Cell-Size Potential $\Phi = c^2 \ln(s/s_0)$

Derives: The cell-size potential takes the universal logarithmic form $\Phi = c^2 \ln(s/s_0)$, fixed up to an additive constant by the shift symmetry and dimensional analysis.

Significance: Identifies the single potential that generates Newtonian gravity, Coulomb’s law, the MOND scale, and the cosmological scale factor as a theorem rather than an ansatz. (*ref:* Supp-11, §sec:universal-potential.)

14.6 No-Singularity Theorem

Derives: The QCT field equations admit no genuine curvature singularities: the cell-size field is bounded below by the finite cell size, and the apparent Schwarzschild singularity is replaced by a smooth interior.

Significance: Resolves the singularity problem of classical general relativity at the structural level rather than via a quantum-gravity hypothesis; black-hole interiors are finite QCT configurations. (*ref:* Supp-11, §sec:no-singularities.)

14.7 Bekenstein–Hawking Entropy from Cell Counting

Derives: The entropy of a QCT black-hole horizon equals one quarter of its area in Planck cells, $S = A/(4\ell_P^2) = k_B A/(4\ell_P^2)$, by direct enumeration of distinguishable cell-boundary configurations on the sonic horizon. Each Planck-area patch of the horizon admits exactly

four distinguishable boundary states, giving $\ln 4$ bits per cell area, which combined with the cell counting reproduces the factor of $1/4$.

Significance: Recovers the Bekenstein–Hawking area law, arguably the deepest known connection between gravity, thermodynamics, and quantum mechanics, and the central target of every quantum-gravity programme, from elementary cell counting on the horizon rather than from semiclassical thermodynamics or string-theoretic D-brane state counting. The entropy is structural rather than statistical-mechanical: it counts actual cell configurations, not microstates of an effective theory. This places QCT alongside the rare quantum-gravity frameworks that reproduce the holographic entropy from microscopic ingredients. (*ref:* Supp-11, §sec:bh-thermo, cell-boundary entropy.)

14.8 Hawking Temperature from Cell-Boundary Excitations

Derives: The horizon temperature $T_H = \hbar c^3 / (8\pi G M k_B)$ follows from the thermal spectrum of cell-boundary excitations at the QCT sonic horizon. The surface gravity in the cell-medium picture, combined with the cell-cycle action quantum $\hbar = 2\varepsilon_0\tau_0$, yields exactly the Hawking formula without invoking quantum field theory on a curved background.

Significance: Recovers Hawking’s 1974 result from cell-medium dynamics alone, paired with the entropy theorem to give a complete QCT account of black-hole thermodynamics from first principles. Where Hawking’s original derivation required QFT on a fixed Schwarzschild background and was viewed as semiclassical, the QCT derivation is structural: T_H is fixed by the same cell discreteness that fixes $\hbar$, with no separate quantum-gravity ansatz. The result preserves unitarity (via cell-count conservation under P1) and so is consistent with information conservation. (*ref:* Supp-11, §sec:bh-thermo, Hawking temperature.)

14.9 UV Finiteness of QCT Quantum Gravity

Derives: The QCT path integral over the cell-size field is UV-finite at all orders, with the cell size providing a natural Planck-scale cutoff.

Significance: Removes the central obstruction of perturbative quantum gravity (non-renormalisable UV divergences) as a structural consequence of finite cell size. (*ref:* Supp-11, §sec:qct-qg, discreteness and UV-finiteness.)

14.10 Theorem U1: QCT Universe as Schwarzschild Interior (Proved within QCT framework)

Derives: The QCT Phase 2 universe (matter domination, $a(t) \propto t^{2/3}$) is the interior of a Schwarzschild black hole of mass $M = 3c^3T/(4G)$, with $r_s = R_H = (3/2)cT = c/H_0$. Eight convergent routes: (1) metric isometry via macroscopic $a(t) \propto t^{2/3}$ from soliton conservation, 3D Gauss-law geometry, and energy conservation (85%); (2) Penrose diagram conformal isomorphism by the identity map, universal in mass and Hubble constant (92%); (3) Phase 2/3 transition shown spacelike via the FRW Bianchi identity (92%); (4) Tolman-Bondi algebraic identity $r_s = R_H$ for any flat FLRW (99%); (5) Misner-Sharp energy-conservation confirmation of $r_s = (3/2)cT$ to 0.05% (99%); (6) spatial flatness $k = 0$ forced by P1 (excludes $k = -1$) and U1 self-consistency (excludes $k = +1$) (88%); (7) structural dissolution of the AMPS firewall paradox since QCT lacks both the factorised QFT Hilbert space and the Bogoliubov interior partner mode (80%); (8) Hawking

temperature $T_H = \hbar c^3 / (8\pi G M k_B)$ from cell-gradient discrete Unruh, without Bogoliubov mixing (88%).

Status: Proved within the QCT framework. Upgraded from proposition (near-theorem 88%) to theorem by TRM-3 closure job `trm-u1-closure`, May 2026, which incorporated the original five-route proof from QCT-Supp-3 (Tolman-Bondi and Misner-Sharp routes had been omitted during the May 14 Supp-11 consolidation).

What remains: Refinement targets only, none structural: explicit cell-Hamiltonian derivation of G in terms of ε_0 and ℓ_P ; formal coarse-graining from the microscopic cell-flow equation to the macroscopic Friedmann equation (the C6 tension); first-principles QCT path-integral derivation of the Hawking spectrum; removal of the mild $U1 \leftrightarrow k = 0$ circularity through an independent QCT critical-density calculation.

Significance: Theorem U1 closes the loop between the fine structure constant, the cell-flow equation, and cosmological evolution. The universe sitting inside a Schwarzschild geometry is not metaphor: it means the same α that fixes atomic structure also fixes the cosmological horizon radius $R_H = r_s$, and the cell-flow equation that drives cosmological expansion is the same flow that governs local soliton dynamics. The fine structure constant and the universe's large-scale structure are one and the same fabric, derived together from a single geometry rather than coordinated through independent inputs. This is the structural unification that makes QCT a single theory rather than a collection of compatible results. The same identification reveals why the periodic table looks the way it does. The fine structure constant $\alpha \approx 1/137$ supports soliton stability for carbon: the six-electron carbon soliton configuration is stable precisely because α sits in the range where the Hopf winding can support the cell count N_c required for a stable six-proton nucleus. A different α produces a different periodic table. The Goldilocks zone for carbon-based life is not a coincidence of initial conditions but a structural consequence of the geometry of D_5 , fixed by the same D_5 Bergman calculation that yields $\alpha^{-1} = 137.036 \dots$ in the first place.

As α varies, whether cosmologically over time through the C6 cell-flow or locally through cell-compression effects, the accessible stable soliton configurations change. The periodic table is therefore not a fixed inventory of elements: it is a function of the local fine structure constant. At reduced α , heavier stable elements become accessible that do not exist at standard α . The growing periodic table is a falsifiable prediction of the framework: the set of stable elements expands as α decreases, and that expansion is calculable from D_5 geometry without further inputs.

The technological consequence is the most far-reaching. If α can be locally altered through cell-compression, as QCT predicts is possible in sufficiently high-density cell configurations, then materials with properties impossible at standard α become constructible within that localised altered-FSC space. Elements and molecular configurations that do not exist under standard laboratory conditions become accessible in altered- α environments, opening a regime of materials science that is structurally predicted by QCT and inaccessible through any framework that treats α as a fixed constant of nature. Advanced material production in localised altered-FSC space is not an exotic speculation; it is the operational reading of Theorem U1 combined with the FSC derivation, and it follows from taking the unification of cosmology and the fine structure constant seriously.

(*ref:* Supp-11, §sec:universe-bh; proofs/theorem-u1-final-v1.md.)

15 Dark Matter and Dark Energy

Source: Supp-12.

15.1 Higher-Charge Hopf Solitons as Dark Matter Candidates

Derives: Hopf solitons of topological charge $Q \geq 2$ are stable, electrically neutral, and decoupled from the standard photon sector; they constitute the QCT candidate for cold dark matter.

Significance: Provides a structural dark-matter candidate within QCT, with stability and electromagnetic darkness as theorems rather than postulated properties. (*ref:* Supp-12, § sec:dm-solitons.)

15.2 Stability of $Q \geq 2$ Solitons

Derives: BPS saturation and topological conservation forbid decay of $Q \geq 2$ Hopf solitons into lower-charge states.

Significance: Explains why dark matter is cosmologically stable on Hubble timescales without imposing a separate stabilising symmetry; topology does the work. (*ref:* Supp-12, § Stability of higher-charge solitons.)

15.3 Flatness Route to $\Omega_\Lambda = 0.704$ (Airtight Algebraic)

Derives: Imposing spatial flatness together with the QCT dark-matter content gives $\Omega_\Lambda = 0.704$ as an algebraic identity, with no free parameters.

Significance: Recovers the observed dark-energy fraction $\Omega_\Lambda \approx 0.69$ from a structural QCT constraint; addresses the cosmological-constant problem from the geometry side rather than the vacuum-energy side. (*ref:* Supp-12, § sec:flatness-route.)

15.4 No Dark Energy: $\rho_\Lambda/\rho_P = e^{-2/\alpha}$

Derives: The apparent vacuum energy density relative to the Planck density is exactly $\rho_\Lambda/\rho_P = e^{-2/\alpha}$, derived from the cell-medium ground-state action evaluated on the D_5 Bergman measure. The exponential suppression by $2/\alpha$ arises from a tunnelling-like overlap between Phase 1 and Phase 2 cell configurations, with the factor of α fixed by the same Wyler identity that produces the fine-structure constant. No independent dark-energy substance, quintessence field, or modified-gravity term is required.

Significance: Solves the 120-orders-of-magnitude cosmological-constant problem with an exponential suppression by $2/\alpha \approx 274$, the only known mechanism that produces the observed value without parameter tuning. “Dark energy” is reidentified as a geometric residue of the cell medium rather than a separate fluid or substance, removing a 70% component of the standard cosmological budget as an independent ingredient. The prediction is closed-form and parameter-free: it matches the observed $\Omega_\Lambda \approx 0.69$ from α alone. (*ref:* Supp-12, § sec:no-de.)

16 Work in Progress

The entries above are results that, at the time of writing, trace cleanly back to P1/P2/P3 through a recorded derivation chain. This section catalogues the next layer: physically significant results that are either established as near-theorems with explicit residual gaps, or open in the sense that the derivation chain exists in outline but a load-bearing step is still incomplete, or closed-out as dead ends after systematic attempts. Including them here is deliberate. A reader who sees only the proved register can mistake the programme for a finished edifice; the live frontier is where confidence levels are being moved, and where the next supplement-level work is happening. Each subsection states the result, the current status, the specific gap that remains, and why closing it matters. Dead ends are catalogued at the end as a reverse map: results which look natural to attempt but which are now known not to work.

16.1 Koide Extremality of $B = \sqrt{2}$ (Near-Theorem 85%)

Derives: The Koide ratio is extremised on the QCT mass-triple constraint surface at $B = \sqrt{2}$, with the extremum coinciding with the measured charged-lepton values to within experimental uncertainty.

Status: Near-theorem 85%.

What remains: The first-principles QCT mechanism for Hilbert–Schmidt equipartition is not yet established; the $CV = 1/\text{Poisson}$ connection is asserted but not derived.

Significance: Explains why nature sits at $K = 2/3$ rather than a generic point on the constraint surface; promotes the Koide identity from coincidence to extremality theorem. (ref: Supp-8, § Koide extremality.)

16.2 MOND Acceleration Scale $a_0 = cH_0/(2\pi)$ (Near-Theorem 75%)

Derives: The MOND crossover acceleration is fixed by the cell-medium relation $a_0 = cH_0/(2\pi)$, derived directly from the universal cell-size potential and the Hubble rate.

Status: Near-theorem 75%.

What remains: Derivation of the exact numerical prefactor $1/(2\pi)$ from first QCT principles; the current result gives the correct order of magnitude but the coefficient is not yet fully pinned.

Significance: Refines the MOND scale from a fitted constant to a calculation in terms of c and H_0 ; pinpoints the deep coincidence Milgrom noticed as a QCT identity.

(ref: Supp-12, § sec:mond.)

16.3 Weinberg Angle $\sin^2 \theta_W$ (Near-Theorem, 90–95%)

Status: Bergman-metric correction closes 94.8% of the gap between the tree-level QCT value $1/(1 + \pi)$ and the measured $\sin^2 \theta_W(M_Z) \approx 0.231$; the residual 0.0005 is consistent with the standard two-loop RGE running from the QCT matching scale.

What remains: The QCT soliton mass spectrum needed to pin the RGE matching scale exactly is not yet available; one supplement file reports an internal inconsistency (0.2327 vs 0.221 across two routes) that must be reconciled.

Significance: Promotes the electroweak mixing angle from a Standard Model input parameter to a D_5 -derived prediction; the only Standard Model coupling not yet at full registry status.

(ref: Supp-6.)

16.4 Muon Mass $m_\mu/m_e = 20\pi^3/3$ (Near-Theorem, 85–96%)

Status: Tree-level closed form $m_\mu/m_e = 20\pi^3/3$ derived from Shilov-boundary path integral at 290 ppm agreement with measurement; the 290 ppm residual is identified as the one-loop QED correction $\alpha_{\text{EM}}/(8\pi)$, reducing the full-formula deviation to -1.2 ppm. Photon-overlap channel equality $I_{\text{bulk}} = I_{\text{boundary}}$ is proved.

What remains: Higher-order CR-curvature corrections on $\check{S}(D_5)$ are not yet computed; the QCT vertex-rule coefficient $\alpha/(8\pi)$ used in the one-loop step does not yet have an independent first-principles derivation.

Significance: Brings the second-generation lepton mass into the same closed-form, parameter-free regime as $m_p/m_e = 6\pi^5$; the muon mass becomes a D_5 boundary-geometry theorem rather than a Yukawa input.

(ref: Supp-4, Supp-8.)

16.5 Koide Relation $K = 2/3$ (Near-Theorem, 75–85%)

Status: The algebraic equivalences $\text{CV}(\sqrt{m}) = 1 \Leftrightarrow K = 2/3 \Leftrightarrow B = \sqrt{2}$ are proved; $B = \sqrt{2}$ is shown to be the extremality value of the positivity-preserving parametrisation of the lepton-mass triple. Two independent geometric routes (B_2 root structure on D_5 ; mean-of-extremes identity) converge on the same value.

What remains: The first-principles QCT mechanism that selects the extremum (candidate routes via Hilbert–Schmidt equipartition or BPS pairing) is established only at ~ 55 – 60% confidence; the parameter $\theta = 2/9$ entering the parametrisation is unexplained at the postulate level.

Significance: Promotes a long-standing empirical regularity of the charged-lepton spectrum from coincidence to derivable identity; together with the muon-mass derivation it closes the charged-lepton hierarchy from QCT alone.

(ref: Supp-8, Supp-10.)

16.6 CMB Spectral Tilt n_s (Reconciliation Required)

Status: One supplement derivation reports $n_s = 1 - 5e^{-\pi^2/2}$ as a closed-form theorem at 0.02% agreement with Planck 2018 (the entry listed in Cosmology above); a separate triage of alternative n_s routes records three systematic derivation attempts that fail. The two records are not yet reconciled in a single source document.

What remains: A unified write-up that states explicitly which route is canonical, why it succeeds, and why the other three are structurally ruled out. Until this reconciliation is written, the n_s entry in the proved register carries an asterisk.

Significance: n_s is among the headline cosmological observables; a clean derivation with a parallel record of failed alternatives strengthens rather than weakens the result, and is the kind of artefact a referee will look for.

(ref: Supp-5.)

16.7 Scale Factor $a \propto t^{2/3}$ (Open: Most Serious Cosmological Gap)

Status: The matter-era scale factor $a \propto t^{2/3}$ is the standard FRW result that QCT must reproduce. A Newton-chain route within QCT delivers $a \propto t^{2/3}$ at near-theorem confidence ($\sim 80\%$); however, the GR8 chain from the QCT cell-resizing field $s(x)$ currently yields $a \propto t^{1/5}$, which gives a universe age of ~ 2.8 Gyr against the observed 13.8 Gyr.

What remains: The disagreement between the Newton-chain route and the $s(r)$ -field route must be resolved; either one route is mis-assembled or the QCT cosmological flow has additional structure not yet captured by the present derivation.

Significance: The most serious open cosmological gap in the programme. Until it is closed, every late-universe prediction (matter-era growth, structure formation, distance ladder, H_0) is conditional on the resolution.

(ref: Supp-5.)

16.8 Cosmological Expansion from P1/P2/P3 Alone (Open)

Status: A direct attempt to derive Hubble expansion from P1/P2/P3 alone finds that P2 drives cell-medium contraction rather than expansion, with no within-postulate route yet identified to flip the sign.

What remains: Either a missing structural ingredient already implicit in P1–P3 must be identified, or a fourth postulate must be added. The present working hypothesis is that the Phase 1→Phase 2 transition supplies the expansion driver structurally, but this is not yet derived.

Significance: The foundational question of cosmology, why the universe expands at all, is the largest single conceptual gap in the QCT derivation chain. Closure would either preserve P1/P2/P3 as a complete postulate set or formally announce the fourth postulate.

(ref: Supp-5.)

16.9 Cold Dark Matter to Baryon Ratio $\Omega_{\text{CDM}}/\Omega_b \approx 5.4$ (Near-Theorem on One Path; Open Otherwise)

Status: A combined route using the baryon-to-photon ratio η , the cosmological constant $\Lambda(\alpha)$, and spatial flatness yields $\Omega_{\text{CDM}}/\Omega_b \approx 5.45$ at near-theorem confidence ($\sim 95\%$). Two alternative direct mechanisms (relic topological fraction, soliton creation rate) are dead ends: the formation-time ratio is $e^{20} \approx 5 \times 10^8$, eight orders of magnitude too large.

What remains: The flatness route requires no new physics, but the question *why* the surviving CDM population sits at exactly the flatness-pinned value rather than the formation-time value is not answered by that route. A first-principles depletion mechanism remains absent.

Significance: The dark-matter abundance is the second-largest cosmological parameter after Ω_Λ ; a structural derivation closes a major degree of freedom of the standard cosmological model.

(ref: Supp-5, Supp-12.)

16.10 SU(3) Colour from SO(5) (Open: Gap in Proton Derivation)

Status: The proton derivation routes the strong sector through SO(5) Yang-Mills via D_5 geometry. An attempt to recover SU(3) colour as a root-theoretic subgroup of SO(5) has been investigated and shown to fail: there is no natural $SU(3) \hookrightarrow SO(5)$ embedding that reproduces the colour content.

What remains: An alternative emergence mechanism for SU(3) (candidates include a Vakulenko-Kapitanski-style topological bound, branching from a larger ambient group, or emergence from soliton-soliton scattering channels) must be identified or the strong-colour structure must be carried by a different QCT object.

Significance: The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is partially recovered ($U(1)$ and $SU(2)$ are in hand); SU(3) is the missing factor. Closure completes the gauge sector.

(ref: Supp-3, Supp-4.)

16.11 Inhomogeneous Maxwell Equations (Open in Some Routes)

Status: The homogeneous Maxwell equations $dF = 0$ are proved as the Hopf Bianchi identity. The inhomogeneous source equations $\partial_\mu F^{\mu\nu} = J^\nu$ are derived from variation of the QCT gauge action on D_5 , but a fully closed first-principles route through every intermediate identification (current J^ν as a cell-medium quantity; gauge action measure-normalisation) is not yet recorded as a single theorem.

What remains: A consolidated derivation that takes the inhomogeneous Maxwell equations from P1/P2/P3 in a single chain, on par with the homogeneous derivation.

Significance: Coulomb's law and the Ampère-Maxwell term are the source-coupled half of electromagnetism; full first-principles parity with the homogeneous side closes the EM derivation chain.

(ref: Supp-1, Supp-6.)

16.12 m_p/m_e Full Radiative Formula (Near-Theorem, 96%)

Status: The Schmid-formula derivation $m_p/m_e = 6\pi^5 = \chi(Q_5) \cdot \pi^5$ via compact/non-compact duality is at 96% confidence; the GK c-function route, which delivers the same prefactor via Gelfand-Kirillov dimension methods, is at 92%. The two-loop coefficient $9/(8\pi)$ is derived; the three-loop α^3 term accounts for the residual 1.3% CODATA deviation.

What remains: A line-by-line write-up of the Schmid 1970 derivation transposed into QCT notation; the GK c-function route requires the same explicit transcription. The isospin-projection argument for the two-loop coefficient is partially slogan and partially derivation.

Significance: The CODATA proton-electron ratio is among the most precisely measured numbers in physics; bringing the QCT prediction from tree-level theorem (already proved at full rigour) to sub-ppm closed-form is the next precision target.

(ref: Supp-4, Supp-8.)

16.13 Two-Soliton Mass Threshold (Near-Theorem, 99%)

Status: The $\lambda = 0$ Plancherel edge of the D_5 representation theory is identified as the two-Rac scattering threshold, with mass gap $\Delta = m_{\min}$. The two-soliton threshold equals

$2m_{\min}$ from the Bogomolny-Faddeev bound on the Plancherel structure.

What remains: Editorial consolidation only; all load-bearing steps are recorded and cross-checked.

Significance: A structural prediction for the QCT scattering sector that complements the T5 mass-gap theorem; the two-particle threshold is fixed by representation theory rather than measured input.

(ref: Supp-3.)

Dead Ends (Closed)

The following derivation paths were attempted systematically and ruled out. They are recorded here because they save future workers the cost of re-attempting them, and because each closure constrains the surviving routes.

Koide Clebsch–Gordan route to $B = \sqrt{2}$. The candidate derivation of the Koide extremality value $B = \sqrt{2}$ from a Clebsch–Gordan coefficient of a K -type structure on D_5 fails: the computed CG coefficients are $c_0 = \sqrt{3}$, $c_1 = 2\sqrt{2}$, $c_2 = \sqrt{15}$, none of which equals $\sqrt{2}$. The Koide value must come from a different structural mechanism (currently routed through extremality of a positivity-preserving parametrisation).

Reversed fine-structure-constant cosmology. The hypothesis that α was smaller at recombination, motivated as a CMB-tilt mechanism, is ruled out: QCT requires α to take its present-day Wyler value at the relevant redshift, and reproducing the observed CMB constraints under a reversed- α history is structurally inconsistent with the Wyler identity. The n_s result must come from D_5 heat-kernel structure, not from α variation.

Cosmological expansion directly from P1/P2/P3. As noted in the open list, attempts to derive Hubble expansion from P1/P2/P3 alone within the present derivation chain fail: P2 drives cell-medium contraction. This is a closure of one route, not of the open problem itself; the expansion driver may still emerge structurally from the Phase 1→Phase 2 transition or from a fourth postulate.

Three systematic n_s alternatives. Three independent attempts to derive the CMB spectral tilt by routes other than the D_5 heat-kernel eigenvalue $5e^{-\pi^2/2}$ have been recorded and shown not to close. They are catalogued for reverse-mapping; the surviving canonical route is the one in the proved register, pending the reconciliation noted above.

Direct soliton-creation route to $\Omega_{\text{CDM}}/\Omega_b$. The formation-time ratio of $Q \geq 2$ to $Q = 1$ Hopf solitons evaluated from QCT counting yields $e^{20} \approx 5 \times 10^8$, eight orders of magnitude above the observed 5.4. The depletion mechanism that brings the surviving abundance down to the flatness-pinned value is not yet identified; the direct creation-rate route is closed.

Early fine-structure-constant routes (FSC-002 through FSC-005). Four sequential derivation attempts of α predating the Wyler programme were documented and ruled out. The closures map the failure modes that the Wyler-route construction had to avoid, and motivate the structural identification of α with the equivariant Wyler measure.

Summary

Group	Entries	Running Total
1. Classical Mechanics	10	10
2. General Relativity	8	18
3. Quantum Mechanics	13	31
4. Electromagnetism	8	39
5. Geometry and Cell Structure	6	45
6. Yang-Mills and the Mass Gap	10	55
7. Particle Masses	5	60
8. Cosmology	20	80
9. Force Unification and Predictions	7	87
10. Chemistry	5	92
11. Extended Particle Masses	11	103
12. Nuclear Physics	5	108
13. Neutrinos	5	113
14. Quantum Gravity	9	122
15. Dark Matter and Dark Energy	4	126
Total (Proved Register)	126	
Work in Progress (near-theorems / open)	14	
Dead Ends (closed)	6	

Each of the 126 entries above traces back, through the derivation chain recorded in the cross-referenced supplement, to the three QCT postulates P1, P2, and P3. No entry depends on a fourth postulate, a tunable parameter, or an unresolved conditional step beyond what is explicitly noted in its source section. The collective scope (from Newton’s second law to the Yang-Mills mass gap, from the photon’s masslessness to the cosmic microwave background spectral tilt, from Coulomb’s law to the baryon asymmetry, from the Kohn–Sham exchange functional to the Bekenstein–Hawking entropy, from quark confinement to the absence of dark energy as a substance) is the central empirical claim of the QCT programme. This registry is the index; the full derivations are in the supplements.

Registry version: 11. *Date:* 2026-05-15.